



SPARROW Assembly & Setup Guide

Solar-Powered Acoustic and Remote Recording Observation Watch

SPARROW Setup Guide

This comprehensive guide is designed to help you seamlessly prepare your SPARROW device for deployment. It's divided into three sections: software setup, hardware setup and sensor setup. By following the step-by-step instructions, you'll ensure a smooth, efficient, and trouble-free setup experience.

Table of Contents

1. Introduction	2
1.1 Open-Source Statement	2
2. Important Safety Information	3
3. Prerequisites	7
3.1 Software Requirements:	7
3.2 Hardware Requirements:	7
4. SOFTWARE SETUP	8
4.1 Starlink Setup and Configuration	8
4.2 Raspberry Pi Setup and Configuration	8
4.2.1 Download and Install the Raspberry Pi Imager	8
4.2.2 Prepare Pi SSD	9
5. Flashing the AudioMoth (OPTIONAL)	12
6. HARDWARE SETUP	15
6.1 Introduction	15
6.2 Component Inventory	15
6.3 SPARROW Hardware Assembly Guide	26
6.3.1 COMPUTE : Assembling the Raspberry Pi 5	26
6.3.2 POWER : Solar Generation, Battery and Network Assembly	31
7 Initial Raspberry Pi Configuration	42
7.1 First Boot and Basic Setup	42
7.2 Registering for a SPARROW Account and Obtaining an Access Key	42
7.3 Transferring and Running the SPARROW Setup Script	43
8. SENSOR SETUP	43
8.1 Configuring ReoLink Wi-Fi Cameras	43
8.2 Peripheral Sensor Assembly (OPTIONAL)	48
9. Field Deployment Tips	50
10. SPARROW Unit Product Specification Sheet	53
11. Weatherproof Enclosure Build Instructions for Tough Conditions (OPTIONAL)	54

1. Introduction

The biodiversity of our planet is rapidly declining. We've likely reached a tipping point where it is crucial to use every tool at our disposal to help preserve what remains¹. **Solar-Powered Acoustic and Remote Recording Observation Watch (SPARROW)**, developed by Microsoft's AI for Good Lab is an AI-powered edge computing solution designed to operate autonomously in the most remote corners of the planet. Solar-powered and equipped with advanced sensors, it collects biodiversity data from camera traps, acoustic monitors, and other environmental detectors that are processed using our most advanced PyTorch-based wildlife AI models on low-energy edge GPUs. The resulting critical data is then transmitted via low-Earth orbit satellites directly to the cloud, allowing researchers to access fresh, actionable insights in real time, no matter where they are.

Printing Recommendation:

For clarity and accuracy, this guide should be printed in **colour**. Many diagrams and component labels **rely on colour coding for correct identification**. Printing in black and white may result in unreadable or misleading instructions, increasing the risk of assembly errors.

1.1 Open-Source Statement

The SPARROW device is an open-source, end-user assembled system. This project is provided as-is, without certification or approval for any jurisdiction.

Because components may vary based on user selection, it is the sole responsibility of the end-user to ensure that:

- All hardware components used comply with applicable local regulatory requirements, including but not limited to:
 - Electrical safety standards for wiring, connectors, and power systems.
 - Radio equipment type approval for any wireless modules or transmitters.
 - Environmental and material compliance (e.g., hazardous substances restrictions).
- The completed assembly meets national and regional certification requirements for the location where the SPARROW unit will be installed.
- Any modifications or substitutions made do not violate local laws or safety regulations.

Failure to ensure compliance may result in legal liability, safety hazards, and operational restrictions. Users should consult relevant regulatory authorities and standards before installation. End-users are strongly advised to seek professional review and guidance before installation or operation to confirm compliance with all applicable laws and standards.

Users follow this guide at their own risk. Microsoft provides these instructions for informational purposes only. Certain steps, such as custom wiring, power configuration, or integration of third-party components, may void manufacturer warranties on individual parts. All risks associated with component selection, installation, and usage rest solely with the end-user. Users are strongly advised to review original component warranties and consult qualified professionals before proceeding.

¹ Hooper, D. U., Adair, E. C., Cardinale, B. J., Byrnes, J. E. K., Hungate, B. A., Matulich, K. L., Gonzalez, A., Duffy, J. E., Gamfeldt, L., & O'Connor, M. I. (2012). A global synthesis reveals biodiversity loss as a major driver of ecosystem change. *Nature*, 486(7401), 105–108. <https://doi.org/10.1038/nature11118>

2. Important Safety Information

CAUTION: Read Assembly & Setup Guide Before Beginning Work

Read this SPARROW Assembly & Setup Guide (“Setup Guide”) in its entirety before beginning work. Pay special attention to the safety warnings and instructions in the Setup Guide.

Building and/or repairing any electronic device can present a risk of electric shock, fire, serious personal injury, death, damage to the device or other property, environmental harm, and/or other hazards. Exercise caution when undertaking the activities described in this Setup Guide.

The activities identified in this Setup Guide should only be undertaken by technically inclined individuals with the knowledge, experience, and specialized tools required to construct an electronic device.

WARNING: Follow all Safety and Health Information

Read and follow all warnings, precautions, handling, and use instructions provided by the manufacturers of hardware components listed in the Hardware Bill of Materials (“BOM”). Failure to follow the manufacturers’ instructions provided for set up, use, and care of the hardware components selected for your build can increase the risk of serious personal injury, death, property damage, and/or environmental harm. You can request additional information and support by contacting the hardware component manufacturer.

In addition, please read and follow the additional safety and health information provided in this Setup Guide. Failure to follow the additional information provided can increase the risk of serious personal injury, death, property damage, and/or environmental harm.

WARNING: Use Personal Protective Equipment

We recommend wearing protective eyewear and gloves when constructing, installing, disassembling, and/or re-assembling a SPARROW unit.

Review and follow the instructions in this Setup Guide when installing, servicing, or inspecting the SPARROW device at heights.

WARNING: Battery Safety

Improper use or handling of devices or their batteries may result in battery failure; electric shock; fire; serious personal injury; death; damage to the device, other property, and/or the surrounding environment.

- Before beginning SPARROW unit construction, ensure your workspace is free of flammable debris or materials, has adequate ventilation, and that you have a fire suppressant device (example: fire blanket, container of sand, Class B fire extinguisher) within easy reach or you are within 20 feet of a fireproof enclosure.
- Keep your workspace clear of debris, extra tools, and sharp objects. Avoid using tools that conduct electricity.

- Exercise caution when using sharp tools near the battery to avoid impacting or poking the battery.
- Do not puncture, impact, strike, bend, or crush the battery. If a battery is impacted, dropped, dented, or punctured during installation, DO NOT proceed. Discard the battery pack and use a new one.
- Do not leave loose screws or parts inside the SPARROW unit.
- Do not heat, puncture, mutilate, or dispose of the SPARROW unit batteries installed in your unit in fire. Do not leave or charge the SPARROW unit batteries in direct sunlight or expose its batteries to temperatures outside the recommended operating range identified by the battery manufacturer for an extended period.
- Do not modify or remanufacture the battery installed for your SPARROW unit; attempt to insert foreign objects into the battery, immerse or expose the battery to water or in other liquids or expose the battery to fire, as this might lead to explosion, or other hazards.
- Protect the SPARROW unit and its batteries from excessive rain, snow, or other types of moisture.
- Ensure you handle, recycle, and/or dispose of used or damaged batteries in accordance with local laws and regulations.

Electrical and Wiring Safety

Always use the proper wire gauge when constructing, installing, or servicing the SPARROW device. Failure to use the proper wire gauge may result in overheating and fire hazards due to undersized wiring, electrical shock or short circuits from improper connection, device malfunction or permanent damage; follow applicable safety, code and compliance standards.

- Follow the wire gauge specifications provided by the component part manufacturer.
- Do not substitute smaller or unapproved wire sizes or types.
- Ensure all connections are secure and insulated to prevent accidental contact.
- Avoid working on electrical installations during adverse weather conditions such as high winds, thunderstorms, rain, ice, or snow.
- We recommend that all electrical wiring-related activities only be conducted by a trained and licensed professional using approved materials.

To help protect your SPARROW unit from environmental hazards and wildlife interference, it is strongly recommended to install any exposed wiring inside approved conduits to prevent physical damage from rodents, birds, or other animals; moisture intrusion that can lead to corrosion or short circuits; or UV degradation of cables in outdoor environments all of which can expose powered wires and pose an electric shock and fire hazard through short circuits.

Always select conduits rated for outdoor use and compliance with local electrical codes. Secure the conduit properly to avoid strain on connections and maintain system integrity.

WARNING: Working at Heights

Do NOT work at heights without proper safety measures and use of personal protective equipment when installing, servicing, or inspecting the SPARROW device at height. Failure to comply may result in falls leading to severe injury or death, damage to equipment, surrounding property, and/or the environment, and violation of workplace safety regulations.

- Always use approved fall protection equipment (harnesses, anchors, guardrails).

- Ensure ladders, scaffolds, and platforms are stable and secure before use.
- Do not work alone—maintain communication with a safety partner.
- Avoid working at heights during adverse weather conditions such as high winds, thunderstorms, rain, ice, or snow.
- Follow all local occupational health and safety standards.

WARNING: Proper Installation

To avoid hazards related to improper device installation, installation must be performed by individuals who have read this Setup Guide and fully understand the installation requirements and environment prior to beginning work.

Consult a professional installer if you do not have the necessary equipment or expertise, or if you are uncertain your proposed mounting environment and mounting solutions will properly support your SPARROW device and not create a hazard, damage, or harm to the installation area or structure.

Safe Ground Anchoring

For ground mounted units, ensure the SPARROW device is securely anchored to stable ground before installation or operation. Improper anchoring may result in device tipping or collapse, causing severe injury, death, or other damage. Insufficient anchoring can lead to system misalignment, reducing performance and reliability and create hazards during high winds, seismic activity, or accidental impact.

- Use approved anchoring hardware and methods as specified by local requirements.
- Verify soil stability and load-bearing capacity before installation.
- Do not install on loose, uneven, eroding, or flood-prone surfaces without adhering to regulations and requirements for this area.
- Inspect anchors regularly for signs of corrosion, loosening, or structural compromise.
- Follow all local building codes and safety regulations for ground-mounted systems.

Failure to properly anchor the unit can lead to catastrophic failure, personal injury or death, and legal non-compliance.

WARNING: Avoid Flammable or Explosive Environments

Do not build, install, or operate the SPARROW unit near flammable materials or within a flammable or explosive environment. Keep all flammable liquids, gases, and solids away from the installation site. Do not install the SPARROW unit in areas with combustible dust, vapors, or fumes. Failure to comply may result in fire or explosion hazards due to the ignition of combustible substances, causing severe injury or death to users, bystanders, and wildlife, property damage, and/or permanent damage to the SPARROW unit and the surrounding environment.

WARNING: Unseen Hazards

Mounting locations can contain unseen hazards or obstacles. Cutting or drilling into an unseen hazard may cause serious personal injury, death, or other harm.

Assess your mounting environment and always make sure there are no unseen hazards associated with the mounting surface prior to drilling and/or cutting. It is the installer's responsibility to locate unseen hazards prior to and to avoid any identified hazards during installation.

WARNING: Ergonomic Lifting

To reduce the risk of lifting related injuries, death, or damage to your SPARROW unit or the environment, use good ergonomic lifting practices during installation, maintenance, and use, including but not limited to:

- Plan ahead. Make sure the lifting team agrees on the plan.
- Determine if you can lift the unit. Is it too heavy or too awkward?
- Decide if you need lifting aid.
- Check your environment for obstructions and slippery surfaces.
- Lift with your legs, not your back.
- Bend at your knees, keeping the back straight.
- Keep the unit close to your body.
- Center your body over the unit. Keep the feet about shoulder width apart.
- Lift straight up smoothly.
- Keep your torso straight; do not twist while lifting or after the load is lifted.
- Set the unit slowly and smoothly into place a straight back; do not let go until the unit is set in place.

WARNING: Solar Panel Glass

Do not install or use the SPARROW unit if any solar panels are cracked, broken, or otherwise damaged. To reduce the risk of personal injury, avoid touching a solar panel that is broken, chipped, or cracked and arrange to have it replaced. Using damaged solar panels may result in electrical hazards, including shock or short circuit, fire risk due to exposed or damaged internal components; and reduced system performance or failure.

WARNING: Do not Modify or Alter the Solar Panel or Battery Terminals

To reduce the risk of fire, electrical shock, causing severe injury or death to users, bystanders, wildlife, the environment, and/or the SPARROW unit, do not modify, tamper with, or attempt to alter the DC terminals of the solar panel array or the terminals of the battery bank used to construct the SPARROW unit.

3. Prerequisites

Before you begin, ensure you have **all the necessary hardware listed in the Bill of Materials (BOM) and software components ready**.

How to Use This Guide

Before starting any assembly, **read each section completely**, including all steps and notes—before taking action. This ensures you understand the sequence, required tools, and safety precautions. Skipping ahead or acting without reviewing the entire section may lead to errors, damage, or safety hazards.

Assembly Time Estimate

Please note that actual build time may vary based on experience. As a guideline, **allow 1.5 hours**.

Power Requirements

Ensure your build space has access to at least **two electrical outlets**. This is necessary to power both the PC (used for flashing and configuration) and the Starlink unit in parallel during setup. Additional outlets may be required for charging tools or other peripherals.

Required Tools

- Monitor with HDMI for Raspberry Pi 5 out
- Zip Ties
- Electric Drill with Conical Drill Step Bit Set (4mm- 35mm)
- Electrical insulation tape and Permanent Marker
- Anti-static wrist strap (optional but recommended)
- Basic Hand Tools:
 - Pliers, Scissors, Angle/Micro Cutting Pliers
 - Wire Strippers
 - Precision Screwdriver Set or Electronics Repair Toolkit
 - Magnifying Glass, Plastic Spudger

3.1 Software Requirements:

- **Host PC for running Raspberry Pi 5 Manager**
- **Internet Connection** (for downloading necessary files and Docker Images)
- **SPARROW Setup Script** (this script can be found in the SPARROW GitHub Repo: <https://github.com/microsoft/sparrow-client>)

3.2 Hardware Requirements:

- **Raspberry Pi 5 8GB**
- **SSD (Solid State Drive) Capacity: Minimum 1TB recommended**
- **Power Supply: Compatible with Raspberry Pi 5**
- **Host PC with internet access**
- **Keyboard, Mouse, and Monitor (for initial setup)**
- **M.2 caddy USB**
- **Mobile Phone**

4. SOFTWARE SETUP

4.1 Starlink Setup and Configuration

Starlink Account Requirements

To ensure uninterrupted connectivity, confirm whether a **pre-existing Starlink account is available** before beginning setup. If no account exists, users must create one and complete activation prior to installation. Include clear instructions for account creation alongside the app installation steps and specify any required credentials or subscription details. This clarification helps avoid delays during deployment and ensures proper configuration of the Starlink unit.

Starlink Setup Note

The Starlink kit includes its own instructions for basic assembly and activation, typically in an outdoor setting. Follow these steps as a **sub-assembly** within the overall process. For clarity, ensure the LED indicator on the Starlink device is lit to confirm it is powered on.

Important: Starlink requires an outdoor connection to establish service. If your assembly location is indoors, plan for an optimal setting where the dish can connect to the sky. If immediate setup is not required, this step can be compartmentalized and completed later in a suitable location.

4.2 Installing Raspberry Pi 5 OS on SSD Using Pi Imager

The Raspberry Pi 5 imager simplifies the process of flashing your Pi device with the appropriate operating system and software packages directly onto the SSD.

4.2 Raspberry Pi Setup and Configuration

4.2.1 Raspberry Pi Imager Installation

1. **Download and Install:** <https://www.raspberrypi.com/software/> (Windows, MacOS or Linux Available).

Raspberry Pi Imager

Raspberry Pi Imager is the quick and easy way to install **Raspberry Pi OS** and other operating systems to a microSD card, ready to use with your Raspberry Pi.

Download and install Raspberry Pi Imager on a computer with an SD card reader. Insert the microSD card you'll use with your Raspberry Pi into the reader and run Raspberry Pi Imager.

[Download for Windows](#)[Read the documentation →](#)[Download for macOS](#)[Download for Linux \(x86_64\)](#)

To install on **Raspberry Pi OS**, type
`sudo apt install rpi-imager`
into a terminal window

4.2.2 Prepare Pi SSD

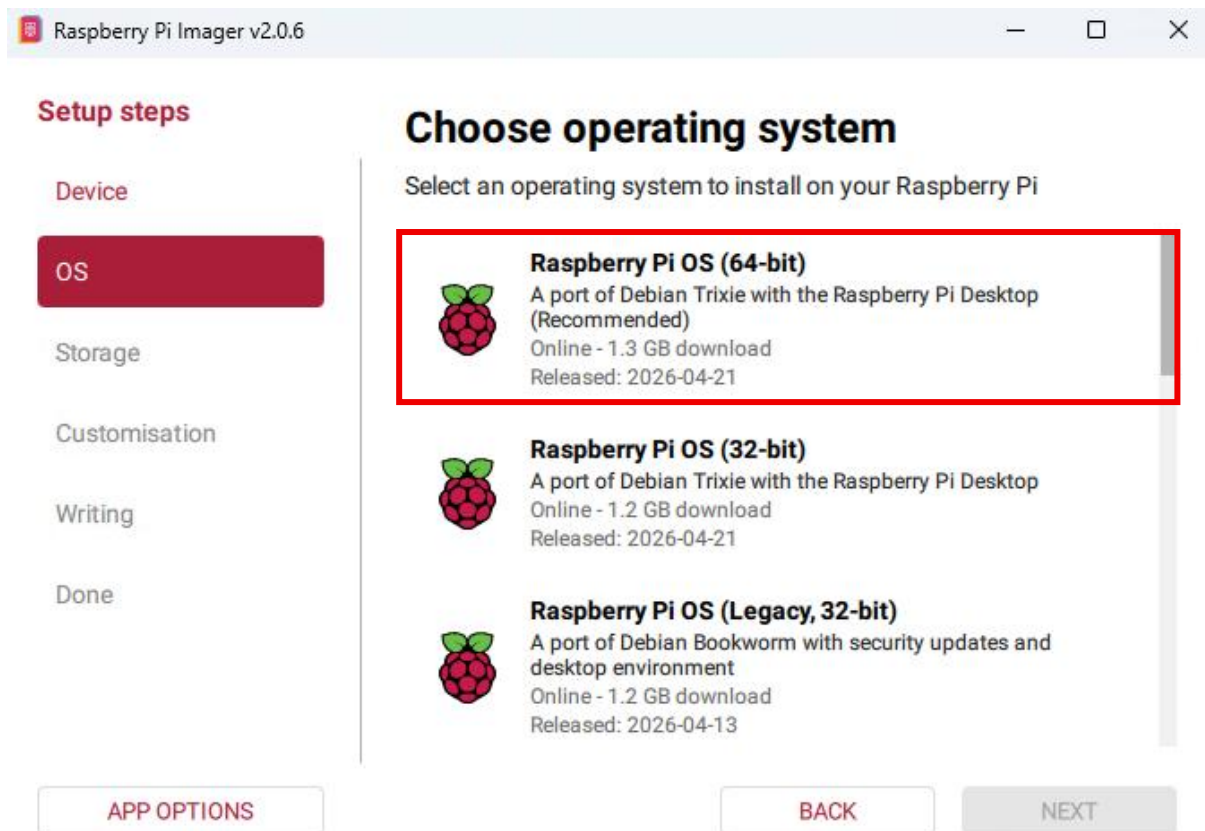
Install the M.2 SSD drive into the USB caddy and connect it to the computer running the Raspberry Pi Imager:



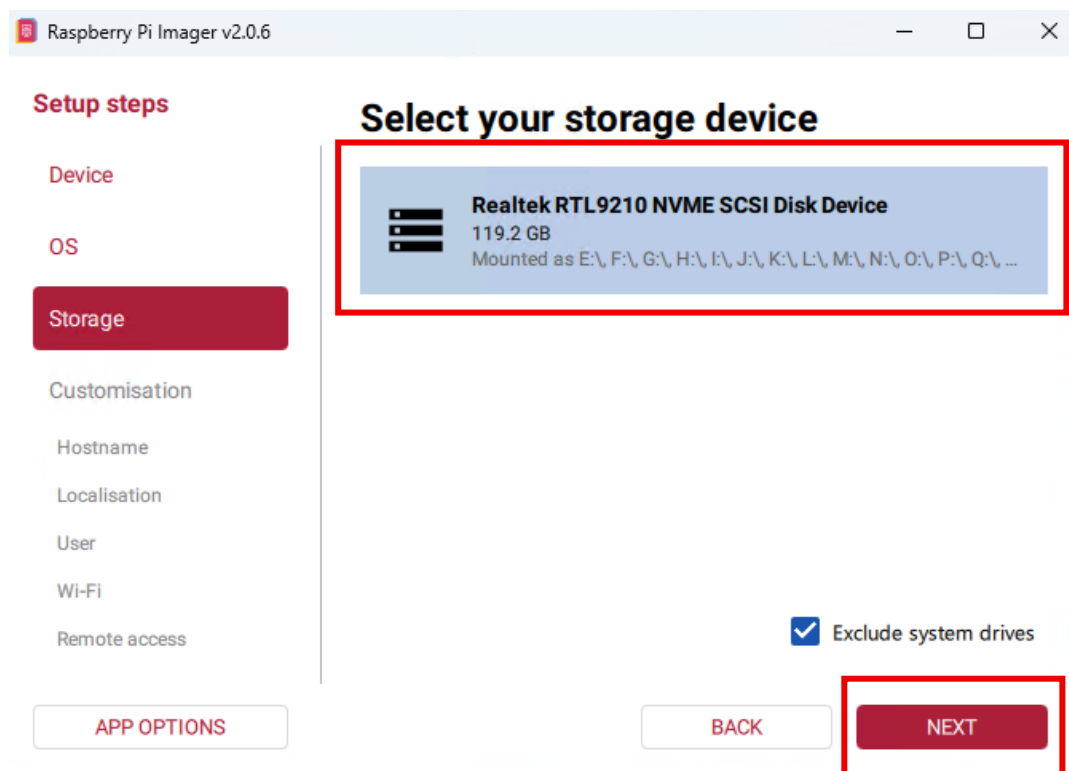
With the USB caddy connected to the computer start the Raspberry Pi Manger. Select Raspberry Pi 5:



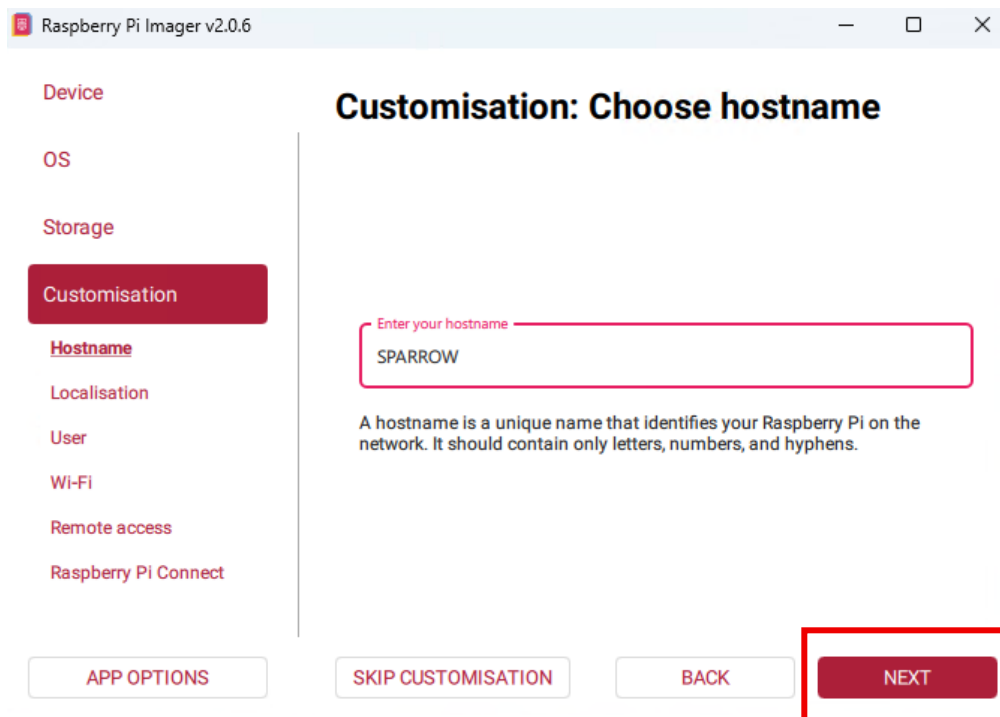
Select Raspberry Pi OS (64-bit) as the image and click next:



Select the M.2 drive and click next:



Enter a host name and click next:



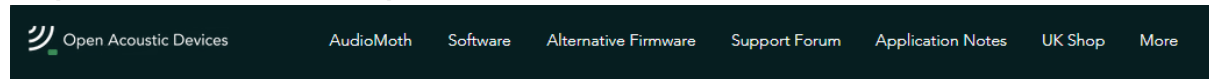
Follow the rest for the configuration prompts. Once the setup is complete you can remove the drive. It is recommended that you configure a Raspberry Pi Connect account to maintain remote access to you SPARROW after deployment.

You can now continue with the hardware build.

Next Steps: You may proceed to the hardware build (Section 6). Note that Section 5 also includes a software-related task (flashing AudioMoth). Ensure you review the entire section before starting, as some activities may be performed in parallel. **Pro Tip:** *While hardware assembly is underway, ancillary software tasks such as flashing AudioMoth can be performed in parallel by another team member, provided all required components and tools are available.*

5. Flashing the AudioMoth (OPTIONAL)

Step 1: Install the USB Microphone Firmware on the host PC



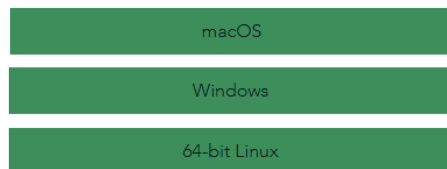
AudioMoth Flash App

The AudioMoth Flash App is used to easily flash your AudioMoth with the latest firmware. Keep your AudioMoth up to date by downloading and applying firmware through this app.

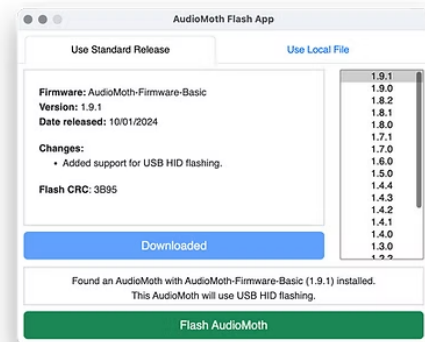
It can also be used to flash devices with your own custom firmware.

Download:

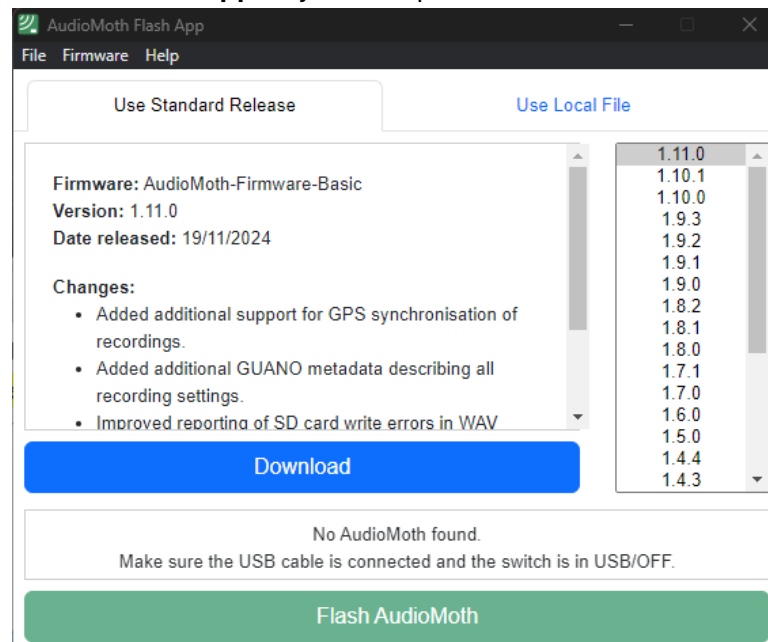
Current version: [1.7.0](#)



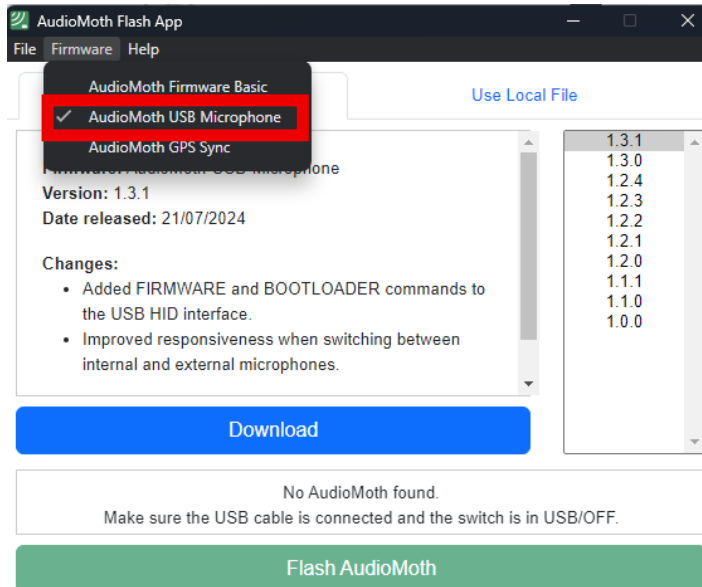
[Additional Linux installation steps](#)



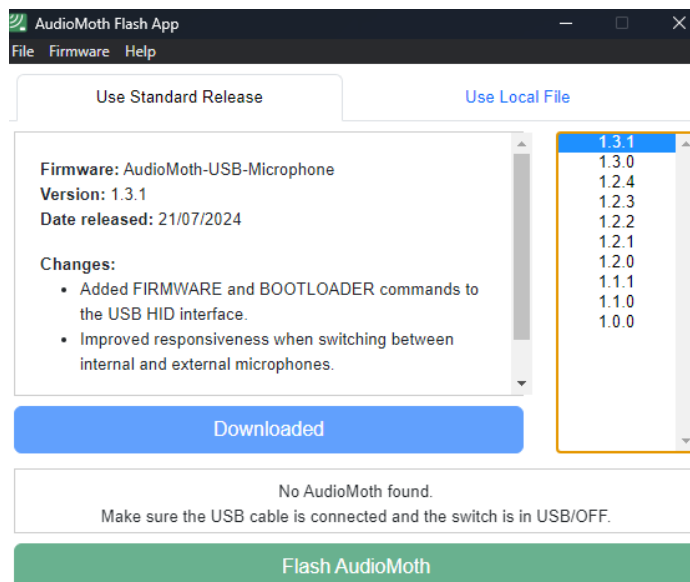
1. **Go to the [AudioMoth website](#)** and download the **AudioMoth Flash App**. The AudioMoth Flash App is used to flash your AudioMoth with the latest firmware.
 - a. Download, Install the AudioMoth Flash App on your computer (you can use Windows, macOS or 64-bit Linux systems).
2. **Open the AudioMoth Flash App** on your computer.



3. **Select the USB Microphone firmware from the Firmware menu.**



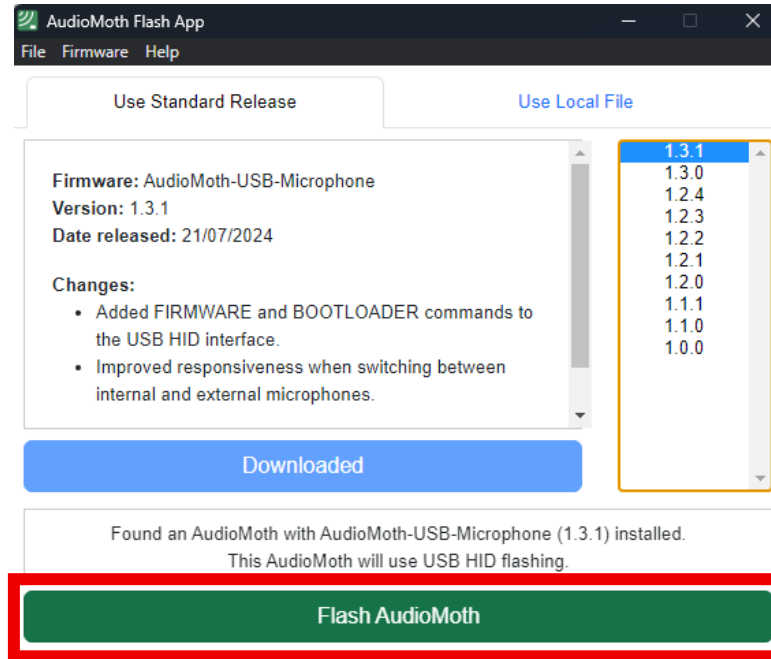
4. **Select the 1.3.1 Firmware version from the list at the right and click the BLUE "Download" Button.**



5. **Connect the AudioMoth board to the computer using the Micro USB port.**
6. **Make sure the micro-USB cable is connected, and white micro-switch in the AudioMoth is set to the USB/OFF position.**



7. **Flash the firmware** by clicking the **GREEN** “Flash AudioMoth ” button.



8. After the flashing process is completed the progress bar will be full and a “*Firmware has been successfully updated.*” message is displayed, at this point you can disconnect the micro-USB cable from the AudioMoth board.
9. Flip the white micro-switch on the AudioMoth from the **USB/OFF position** to the **CUSTOM position**.
10. **Your AudioMoth is now ready for deployment.**

6. HARDWARE SETUP

6.1 Introduction

The SPARROW device can be divided into 3 main component systems:

- **COMPUTE:** Includes the solar controller, edge device and associated devices and the low earth satellite networking solution.
- **POWER:** Solar panels or renewable energy generation components, battery which can be configured differently depending on the deployment scenario.
- **SENSOR:** Includes the Peripheral Sensors (Cameras, Microphones, External Sensors, etc.) that surround the SPARROW unit.

The following steps will guide you through the assembly process of a standard SPARROW unit using solar energy. The guide will cover all necessary components and describe how the components interact with each other. If you follow the steps outlined in this guide, you should have a fully functional SPARROW unit ready for deployment.

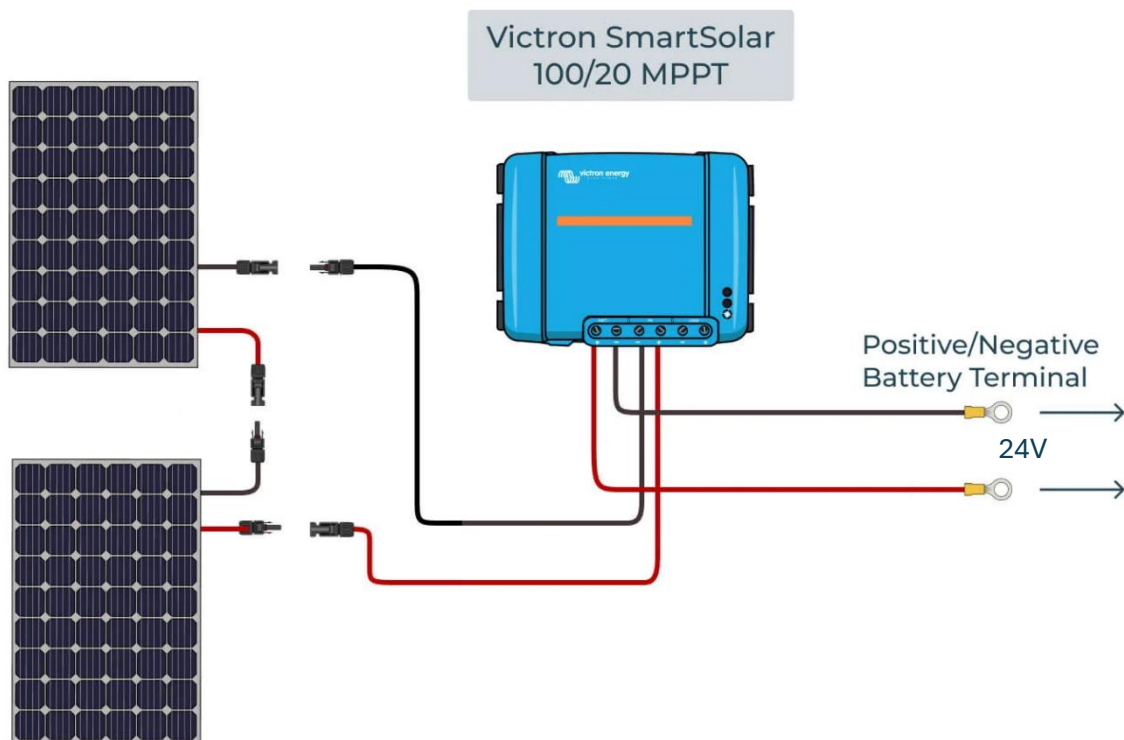
6.2 Component Inventory

The following components highlight each of the required parts to build a full SPARROW unit. This guide provides a brief explanation and description of the function of each component and how it works in the overall system.

You can download the complete SPARROW Bill of Materials from: aka.ms/sparrowbom

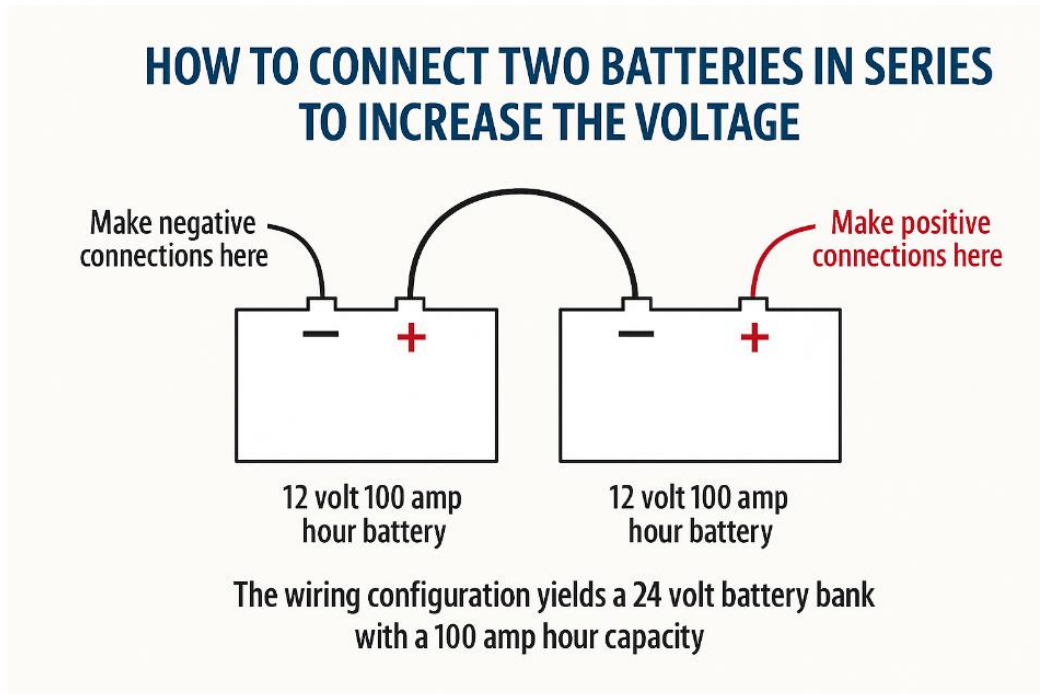
Solar Panel Array:

The solar panels should be connected in a 24 Volt array (2x 12v solar panels working in series).

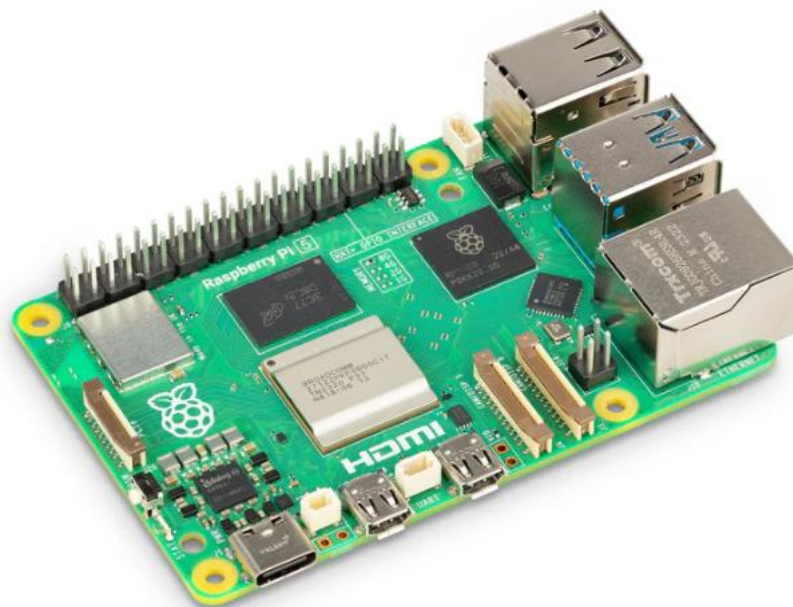


Batteries:

You can use either LiFePO4 or AGM batteries. Since the system works in a 24V configuration, you can use either a single 24V battery or (at least) a pair of 12V batteries running in series as shown in the diagram below.



Edge Computing Device:



The SPARROW unit uses a Raspberry Pi 5 (8GB RAM version). This component runs all the required software services including the AI models. All the sensors directly interface with the Raspberry Pi to provide real-time environmental readings.

Solar Charge Controller:



The solar controller is responsible for maintaining and charging the batteries. It also provides the required voltage to the entire SPARROW system. It also manages the various power profiles to ensure continuous operation and recovery in the event of power loss.

Solar Charge Controller USB Data Cable:



The solar controller USB data cable is responsible for interfacing the solar charge controller with the main edge computing device. This enables the system to remotely monitor system voltages charging profiles, solar generation values, and overall health of the power system.

Pi 3 Click Shield:



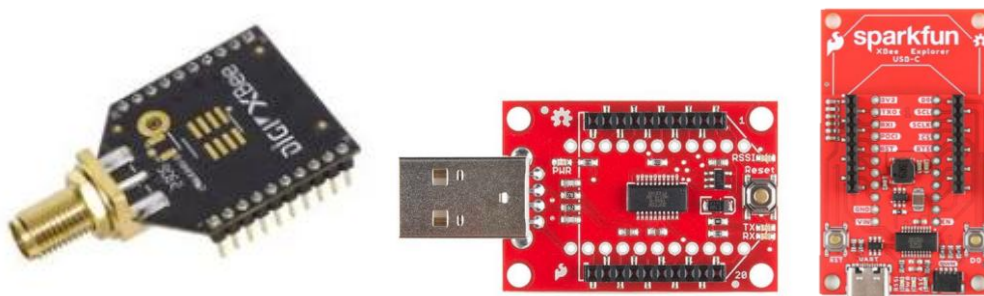
The Pi 3 Click Shield converts the 40 pin GPIO 40-pin expansion header into two female MicroBUS connectors.

Environment 3 Click Board (OPTIONAL):



Environment 3 Click is a four-in-one environmental high-linearity/high-accuracy pressure, humidity, and temperature sensor. The BME688 can detect Volatile Organic Compounds (VOCs), Volatile Sulfur Compounds (VSCs), and other gases such as carbon monoxide and hydrogen in part per billion (ppb) range. It provides absolute temperature accuracy, typical of $\pm 1^{\circ}\text{C}$, and best performance when operated within the pressure, temperature, and humidity range of 300-110hPa, 0-65°C, and 10-90%RH. This is the sensor that is used to monitor atmospherics outside the main SPARROW case/housing. This board is optional, but is supported out of the box.

XBee RF with UART to USB adaptor:



XBee 898/900 is a compact XBee RF module add-on board for long range low-power wireless communication applications. The module can run either a proprietary DigiMesh or point-to-multipoint networking protocol with reliable long range performance and, data redundancy. The **UART to USB adaptor** can be either USB A or USB C, any of these will work.

915MHz LoRa XBee Antenna:



High-gain 915 MHz XBee/LoRa antenna designed to extend wireless communication range and improve signal reliability in outdoor deployments.

M.2 SSD:



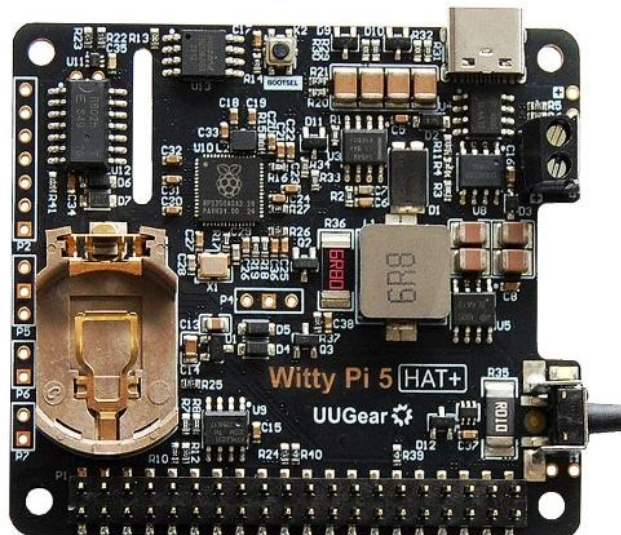
The M.2 NVMe SSD is the main compact, high-speed storage device that plugs directly into the NVMe Base board. The main operating system as well as all data logged will be stored locally to this device.

Relay 3 Click:



Relay click 3 is a dual relay Click board, featuring two single-pole double-throw relays which can be operated by output pins of the host microcontroller (MCU). This board can control a wide range of power applications. The two SRD-5VDC-SL-C relays used on this Click board are hermetically sealed relays which require 5V across their coils while consuming about 0.4W while active. They allow up to 28VDC across the connected load, while conducting up to 7A. The board is used as a remote ON/OFF switch to manage power to Starlink.

Witty Pi 5:



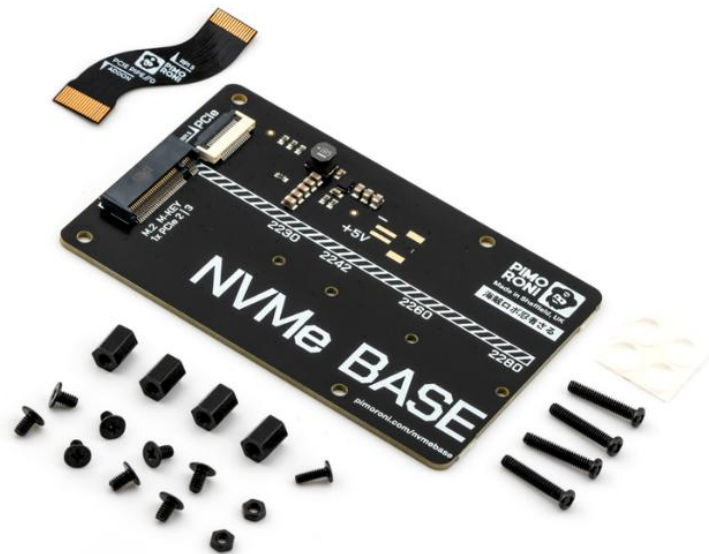
The Witty Pi 5 HAT+ is essentially a smart power-management board that sits on top of a Raspberry Pi. Its main job is to turn the Pi on and off automatically while restoring power to the Pi in the event of a power failure. The Witty Pi 5 doesn't just schedule power; it also contains a wide-input DC power regulator. It accepts roughly 6V–30V DC input and converts that down to the stable 5V required by the Raspberry Pi.

Starlink Power Socket Connector (OPTIONAL):



Female 5.5x2.1mm Female DC Adapter. This is used instead of the assembled orange cable shown above, if you want to DIY build a Starlink female power connector.

NVMe Base:



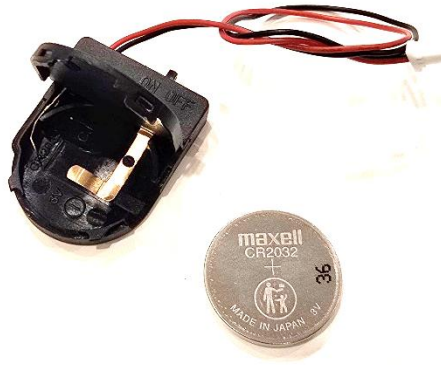
The NVMe Base for Raspberry Pi 5 is a compact expansion board that connects directly to the Raspberry Pi 5's PCIe interface, allowing high-speed NVMe SSDs to be used instead of a microSD card. This provides significantly faster storage performance, improved reliability, and the ability to boot the operating system directly from an SSD.

WAGO Connectors:



A WAGO connector is a tool-free electrical connector that uses a spring clamp or lever mechanism to secure wires. It's widely used because it's **fast, safe, and reusable**, eliminating the need for twisting or crimping wires.

Raspberry Pi RTC Battery Case and CR2032 Battery:



The Raspberry Pi RTC (Real-Time Clock) battery keeps the clock running when the Pi is powered off, preserving the correct date and time between reboots. This allows the system to maintain accurate timestamps for logs, files, and scheduled tasks even without an internet connection.

Electrical Junction Box (OPTIONAL):



In cases where a higher weatherproofing rating might be needed (such as rainforest installations or other equally tough extreme environments) these electrical junction boxes are IP65 rated, or NEMA 4(x) equivalent rating, making it suitable for outdoor applications and able to withstand rain, dust, and other outdoor elements.

Wire Spool:



18–20 AWG (you can go with any gauge between 18 and 20 AWG) black and red wire spool, this is the standard wiring we'll use throughout the SPARROW unit build for all power connections.

Cable Glands (OPTIONAL):



Assorted waterproof cable glands (50 pcs, adjustable 3–16 mm) with gaskets, used throughout the SPARROW unit build for secure, weatherproof cable routing.

Solar Panel Wire with PV004 Connectors (OPTIONAL):



Used depending on how far the solar panels are from the compute unit, these are used for connecting the solar panels PV004 connectors to the solar charge controller.

From Battery to SAE Connector Wire Set:



This is the wire set that will connect to the battery terminals. It features a quick disconnect SAE connector and includes a Positive ATC Style Fuse Holder with 30A Fuse and 2pcs extra fuses.

Batteries SAE Connector to PV004 Connector - Solar Charge Controller Wire Set:



This cable will connect the battery to the solar controller. It includes the necessary polarity inverter adaptor to maintain the polarity from the battery cables.

STARLINK Weatherproof Connector:



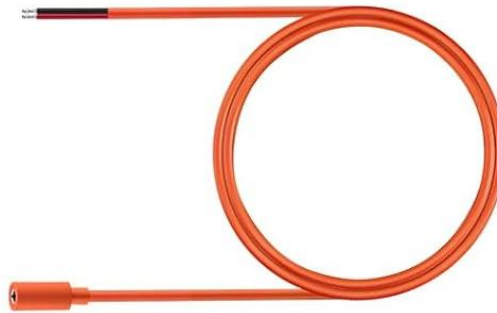
This Starlink Adapter includes a waterproof kit for standard Ethernet cables, designed to connect with a female RJ45 adapter. Once connected, both the female and male ends are fully waterproof, protecting your network components from moisture and dirt, ensuring reliable connections.

AudioMoth DEV Kit and Case (OPTIONAL):



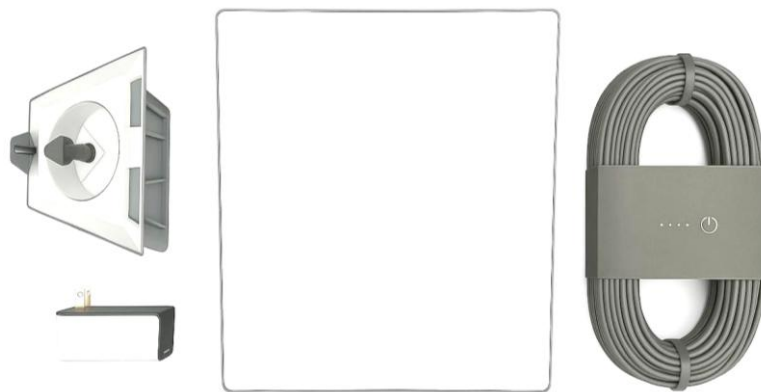
The AudioMoth Dev Kit and Case is a waterproof injection moulded enclosure for AudioMoth Dev. The underside of the case has an M16 cap that can be interchanged to fit external cable glands.

Starlink Power Socket Cable (OPTIONAL):



This DC Power Cable features a 5.5mm x 2.1mm DC female plug used for connecting the Starlink Power Cable. You can use this or the following Starlink Power Socket Connector and build your own Starlink power cable.

Starlink Mini Kit



Starlink Mini is a compact, portable satellite internet kit with an integrated Wi-Fi router, delivering high-speed, low-latency connectivity anywhere with minimal power consumption. Included in this kit is the Starlink DC power cable used in this system.

Large Outdoor Weatherproof Extension Cord Cover Box:



Weatherproof electrical connection enclosure designed to protect extension cords, adapters, and outdoor wiring from rain, dust, snow, and accidental contact. Its reinforced IP54-rated construction, sealed cable ports, and locking clips provide secure, long-term protection for power connections in demanding indoor and outdoor environments.

Dual Lock or Velcro Mounting Tape:



Reusable hook-and-loop fasteners are used to secure sensors, and equipment while allowing quick removal or repositioning. Provides a simple, non-permanent mounting solution that reduces vibration and protects surfaces from damage.

CAT 6 Outdoor Rated Ethernet Cable:



Outdoor-rated Cat6A Ethernet cable designed for high-speed network connectivity up to 10 Gbps. Built with UV- and weather-resistant insulation for reliable data transmission in outdoor and harsh environmental conditions.

Micro USB to USB A cable for AUDIOMOTH (OPTIONAL):



Micro USB cable used for connecting the AudioMoth recorder to the SPARROW main unit.

IMPORTANT: An average length of at least 3' is recommended.

Monitor or portable monitor with a Display Port Cable or Adaptor



A portable monitor is necessary for effortless connection to the Raspberry Pi 5.

6.3 SPARROW Hardware Assembly Guide

6.3.1 **COMPUTE**: Assembling the Raspberry Pi 5

1. Gather Required Tools and Components

- M.2 NVMe SSD (PCIe Gen 3 recommended; sizes 1TB or larger are recommended)
- Raspberry Pi 5
- Small Phillips screwdriver (usually size #0 or #1)
- Anti-static wrist strap (optional but recommended)
- Mounting screw (included with the Raspberry Pi 5 8GB version kit)

2. Power Down and Prepare the Workspace

- Shut down the Raspberry Pi 5 completely.
- Disconnect the power supply and all peripherals (keyboard, mouse, HDMI, etc.).
- Work on an anti-static mat or non-conductive surface to avoid electrostatic discharge.

Mounting the Pi 5 with Velcro to the internal Backplate:

Tools & Materials you will need for this section:

- Raspberry Pi 5
- Weatherproof Outdoor Extension Cord Cover Box or Electrical junction box with perforated backplate
- Heavy-duty zip ties (Optional)
- Drill (if holes need enlarging) (Optional)
- Marker or pencil

Shuttle Click Boards + Sensors Installation + Solar Control Unit

Tools & Materials

- Raspberry Pi 5
- Pi 3 Click Shield (40-pin GPIO compatible)
- NVMe Base
- Witty Pi 5
- Sensors (to connect via Shuttle Click BUS Boards):
 - Environment 3 Click (Optional)
 - Relay 3 Click
- Anti-Static Wrist Strap (optional but recommended)

Step by Step Guide:

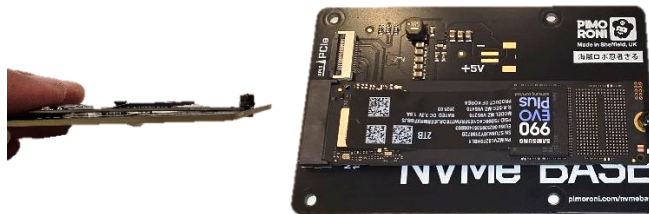
1. Prepare the workspace.
2. Insert your imaged SSD into the NVMe base and attach the data ribbon and four mounting screws as described by the following sequence:
 - a) Check you have all the parts shown below:



- b) Start by mounting the SSD into the board by gently pushing it into the M.2 socket:



- c) Use the thinnest screw together with both nuts to lock the SSD in place:



- d) Install the data PCIe ribbon into the socket. Install it respecting the correct data flow direction, the "ADDON" text end should connects to the NVMe BASE Board while the Pi 5 will be used to connect it to the Pi 5 unit as shown below:



- e) Install the stand-up screws and nuts as described below, this will allow to secure the board to the Pi 5 Unit:



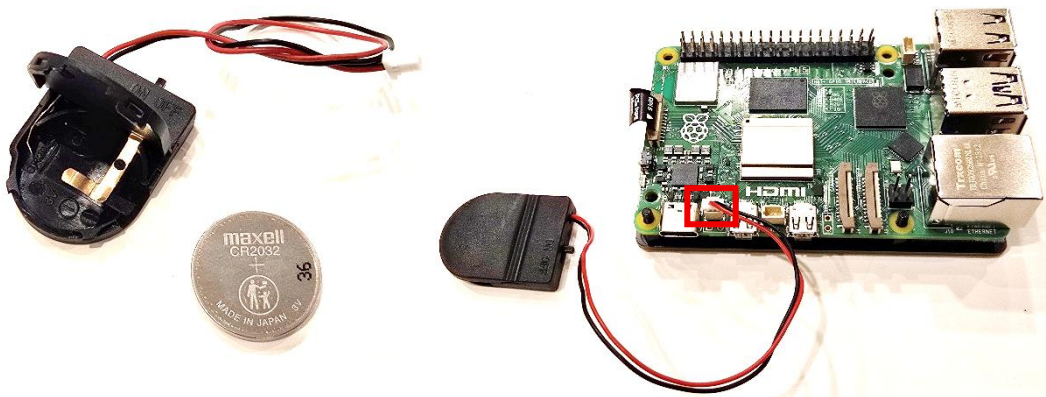
f) This is how the fully assembled NVMe board should look like:



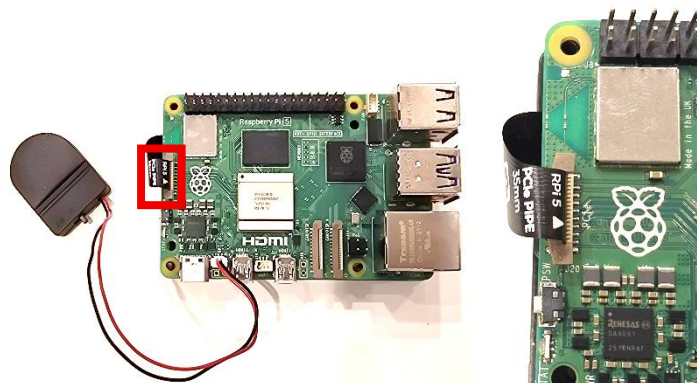
3. Attach Pi 5 to the NVMe base board by placing the Pi 5 over the screws:



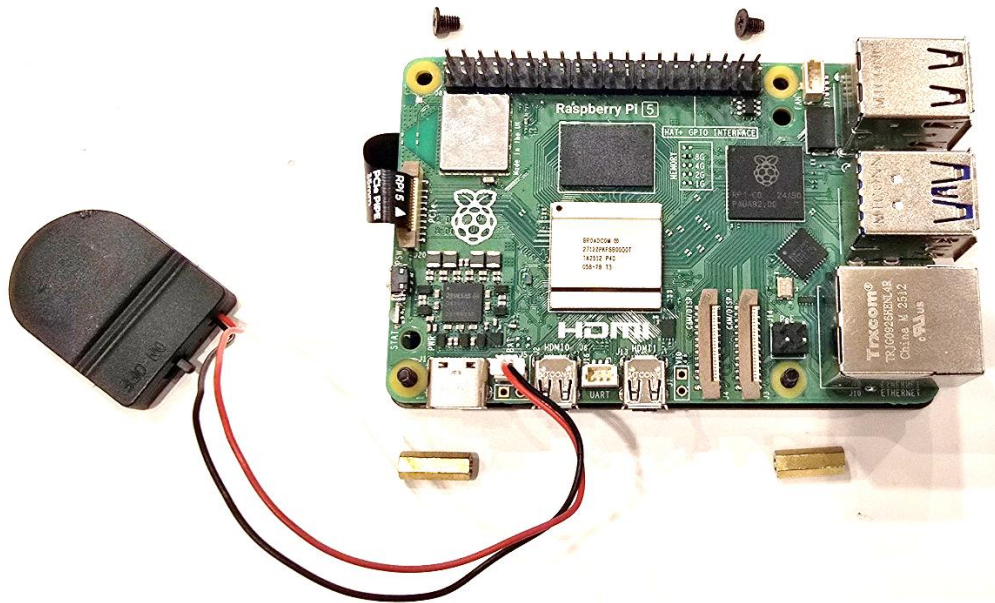
4. Insert the CR2032 battery into the battery housing and connect the RTC battery to the Pi 5 RTC battery JST socket:



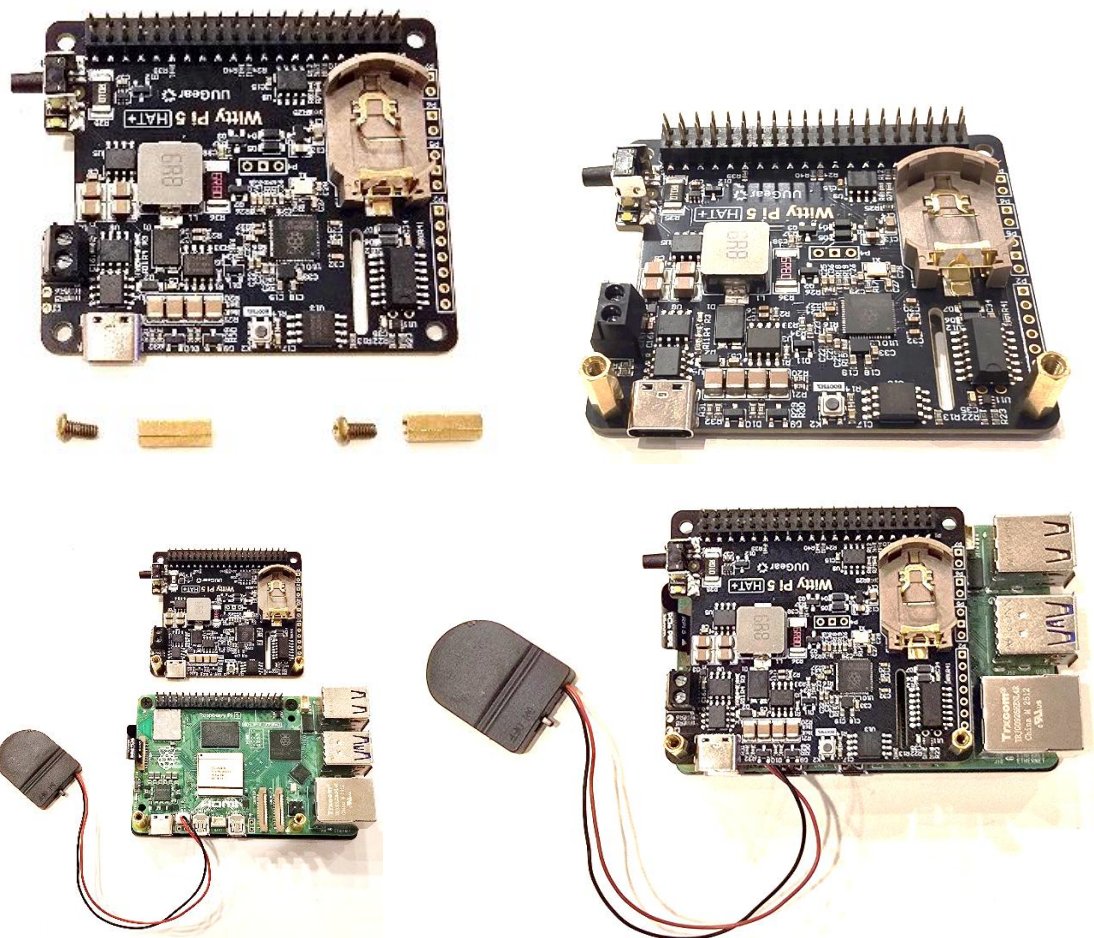
5. Connect the NVMe data ribbon:



6. Install the mounting screws and 2 of the brass stand-off nuts that come with the Pi 3 Click Shield kit. Install them as per the following picture (short screws on the top next to the GPIO pinout, stand-off nuts on the bottom):



7. Attach mounting screws and 2 of the brass stand-off nuts and screws that come with the Pi 3 Click Shield kit to the Witty Pi 5 hat and mount it to the Pi 5:

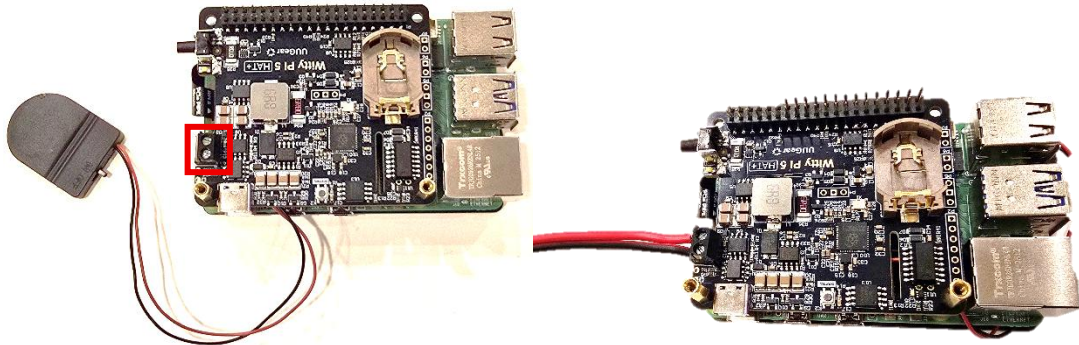


8. Cut a 5 inch (15 cm) long piece of the **RED** and **BLACK** wire and strip the wires ends to expose ½ inch (around 1cm) of the core filaments.

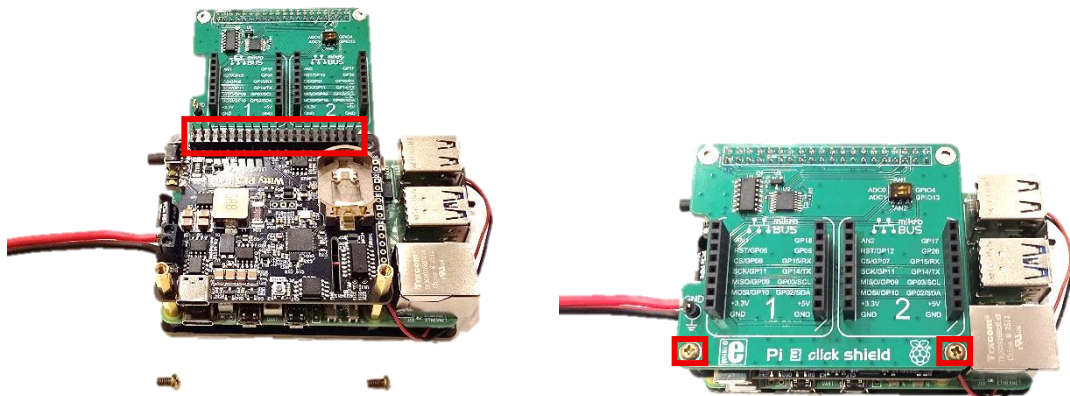


9. Connect the power cable to the Witty Pi 5 board terminals marked in **RED** below. Follow the polarity as described in the board: Connect the **RED** wire to the **POSITIVE (+)** and the **BLACK** wire to the **NEGATIVE (-)**.

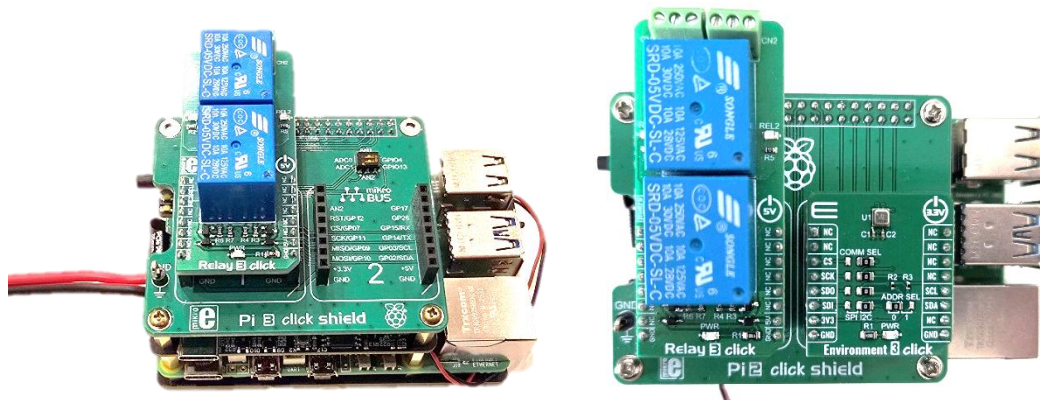
ProTip: You can tuck the RTC battery between the NVMe Board and the Pi 5.



10. Locate the 40-pin GPIO male header on the Raspberry Pi 5 as shown in **RED** below and attach the Pi 3 Click shield. Once installed secure with the two screws:



Install the Relay click 3 Board on the number 1 socket:



11. **OPTIONAL:** At this point you can install the Environment 3 Click Board on the number 2 socket:

6.3.2 POWER: Solar Generation, Battery and Network Assembly

1. Cut a 5 inch (15 cm) long piece of the **RED** and **BLACK** wire and strip the wires ends to expose ½ inch (around 1cm) of the core filaments.
2. Connect the **RED** wire to the **POSITIVE (+)** and the **BLACK** wire to the **NEGATIVE (-)** Load output of the solar controller by unscrewing the Flathead screw of the terminal and inserting the stripped cable end into it and then tightening back, firmly securing the cable in place.



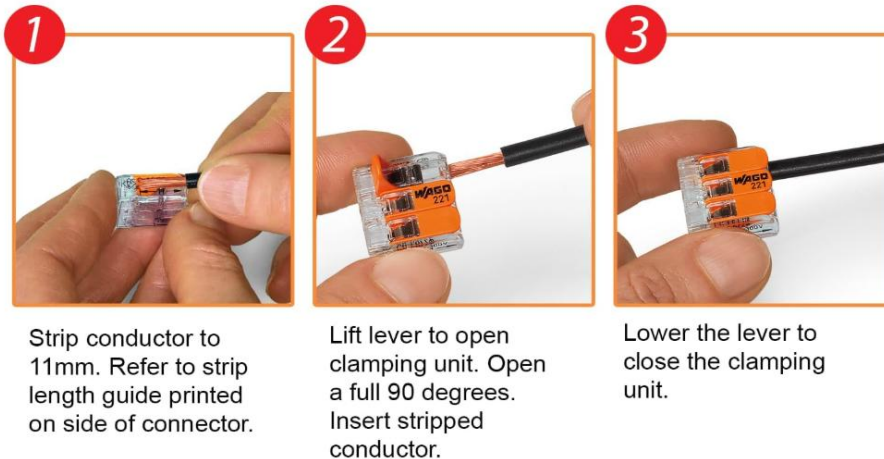
3.

Tip: Twisting the wire ends helps to keep the filament strands from fraying.

4. Attach a 3 port 221 WAGO connector to each wire tip: one connector for the **RED** wire and a different connector for the **BLACK** wire.

Do NOT insert both wires into the same connector under any circumstance as this will create a SHORT and damage the solar controller.

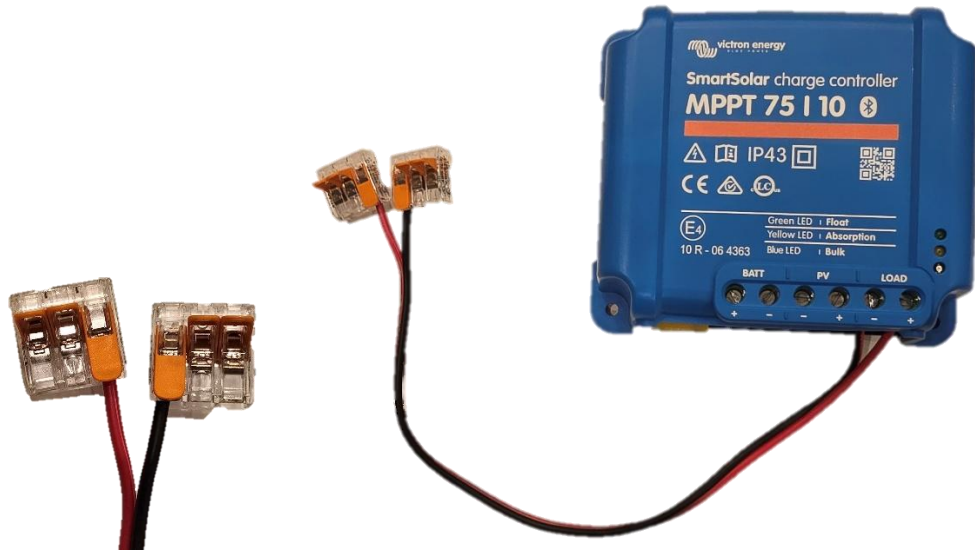
How to use WAGO connectors?



WAGO connectors are very easy to use:

- a. Ensure the copper conductor is clean and undamaged.
- b. Lift the lever on the Wago connector to open the clamping point.
- c. Push the stripped end fully into the connector slot until it stops and make sure no bare copper is exposed outside the connector.
- d. Press the lever down until it locks in place. This clamps the wire securely.
- e. Gently tug the wire to ensure it's firmly held.
- f. Double-check that the insulation is flush with the connector edge.

* To remove a wire simply lift the lever and pull the wire out.

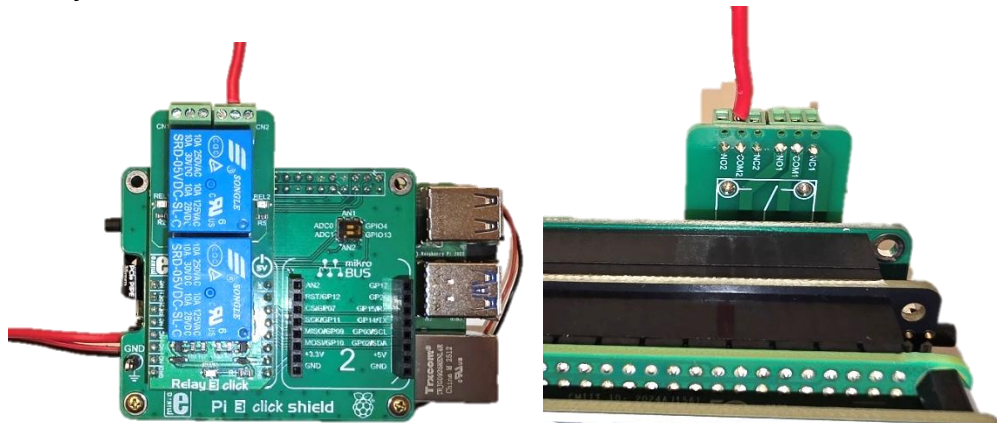


5. We will now start wiring the Relay Click 3 Board. Cut a 4 inch (10 cm) long piece of **RED** wire and strip the wires ends to expose ¼ inch (around 0.5cm) of the core filaments on one end, and ½ inch (around 1cm) on the other.



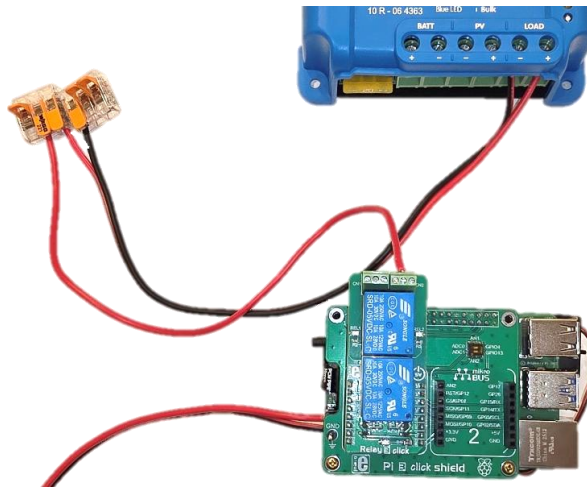
6. Connect the short ¼ inch (around 0.5cm) stripped end to the **COM 2** terminal of one the relay output connectors in the Relay Click 3 Board. You can use either one. To do this simply unscrew the screw for the **COM 2** port and push the stripped end fully into the connector slot until it stops and make sure no bare copper is exposed outside the connector.

Pro Tip: In the case the bare twisted cable core diameter is too big to fully fit inside the Relay Click 3 Board relay output port, you may carefully trim some filaments until the cable does fit. As a rule of thumb try to remove the minimum possible amount of material necessary to make it fit.

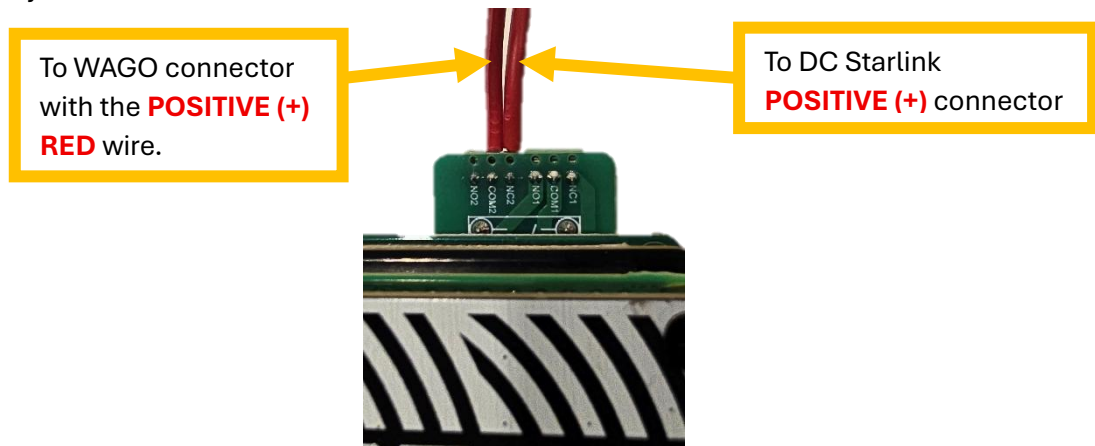


7. Connect the ½ inch (around 1cm) stripped end to a free port of the WAGO connector that contains the **RED POSITIVE (+)** wire we installed previously.

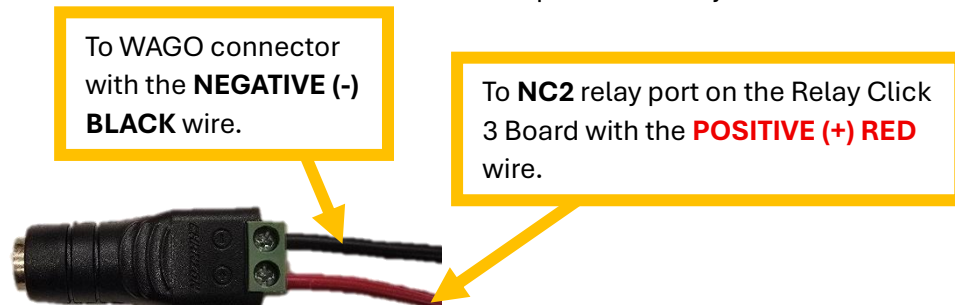
The result should look like this:



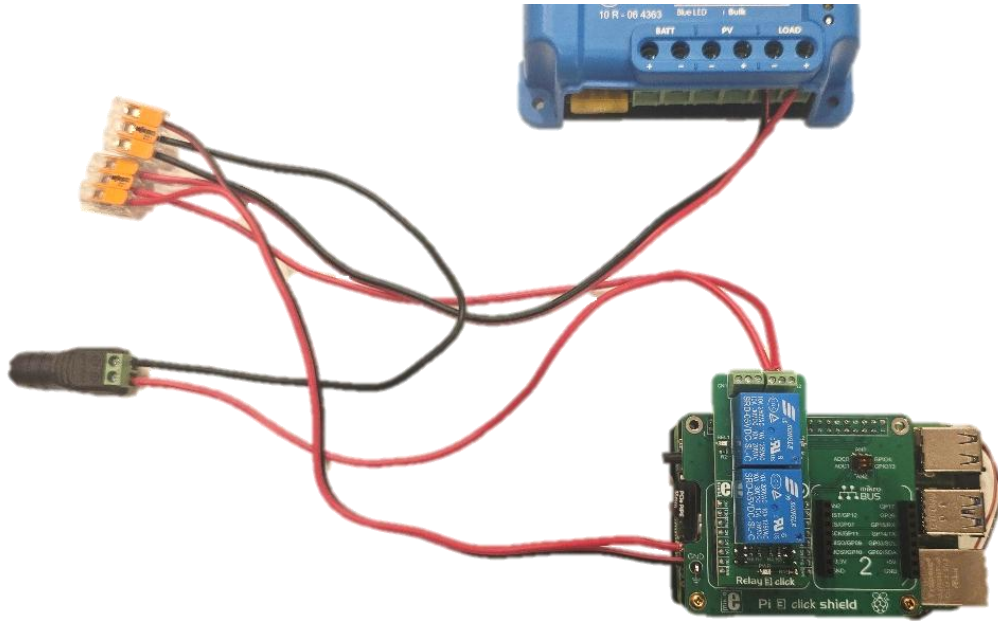
8. Cut a 5 inch (15 cm) long piece of the **RED** and **BLACK** wire and strip the wires ends to expose ¼ inch (around 0.5cm) of the core filaments on one end, and ½ inch (around 1cm) on the other.
9. Connect the stripped ¼ inch (around 0.5cm) **RED** wire end to the **NC 2** relay port on the Relay Click 3 Board. The result should look like this:



10. Connect the ½ inch (around 1cm) stripped core end of the **BLACK** wire to the WAGO connector that had the **NEGATIVE (-) BLACK** wire we had previously installed.
11. Connect the other free end of the **RED** wire to the **POSITIVE (+)** terminal of the DC Starlink power female jack and the other **BLACK** wire ¼ inch (around 0.5cm) stripped end to the **NEGATIVE** terminal of the DC Starlink power female jack.



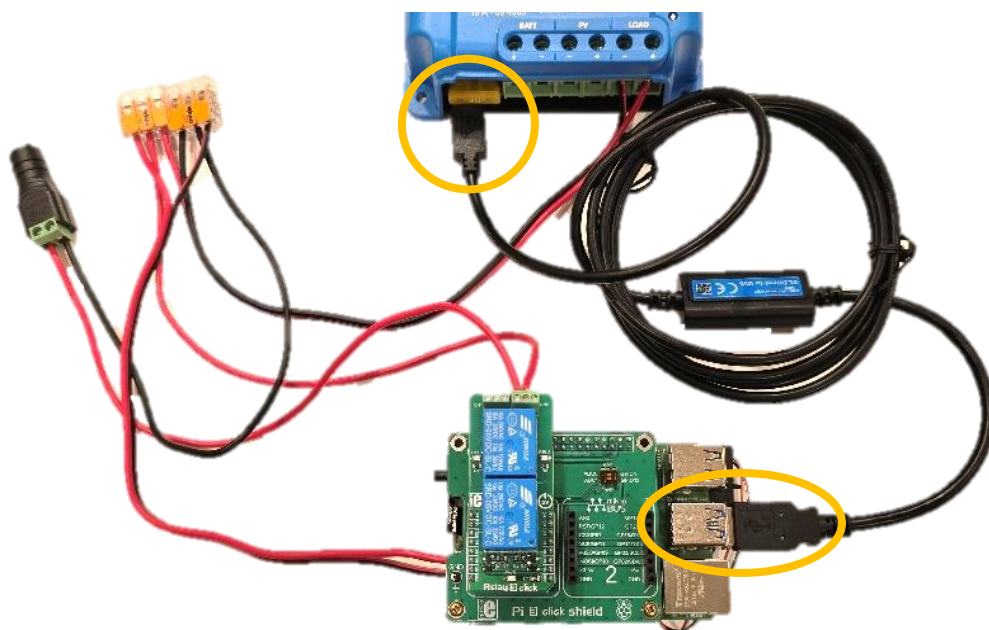
12. The result should look like this:



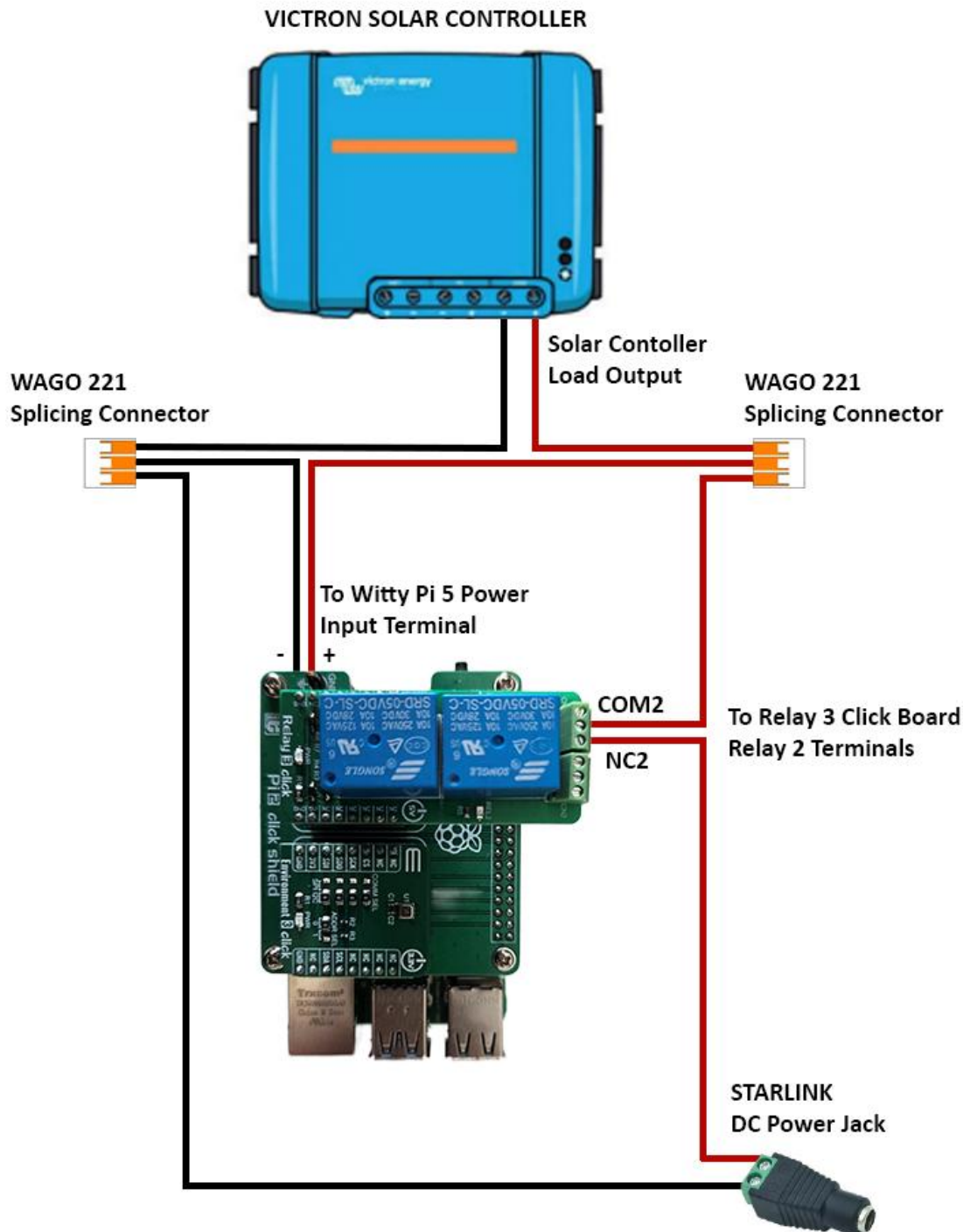
13. Now, install the Victron Energy USB bus cable going from the front left white port of the Victron solar controller to any free USB port on the Raspberry Pi (first remove the Black jumper that comes by default on the white port before installing the white connector end of the USB BUS cable to the solar controller):



14. At this point, the assembled system should look like this:



15. This is how the assembly should look like at this point:



16. Prepare the battery and solar cable leads

Before installation, decide how much cable length is needed between the main compute unit, the solar panels, and the battery. The cable assembly can be cut to match the installation layout.



We recommend cutting the cable assembly in half. This gives enough length on both sides to position the solar panels and battery with some flexibility during installation.

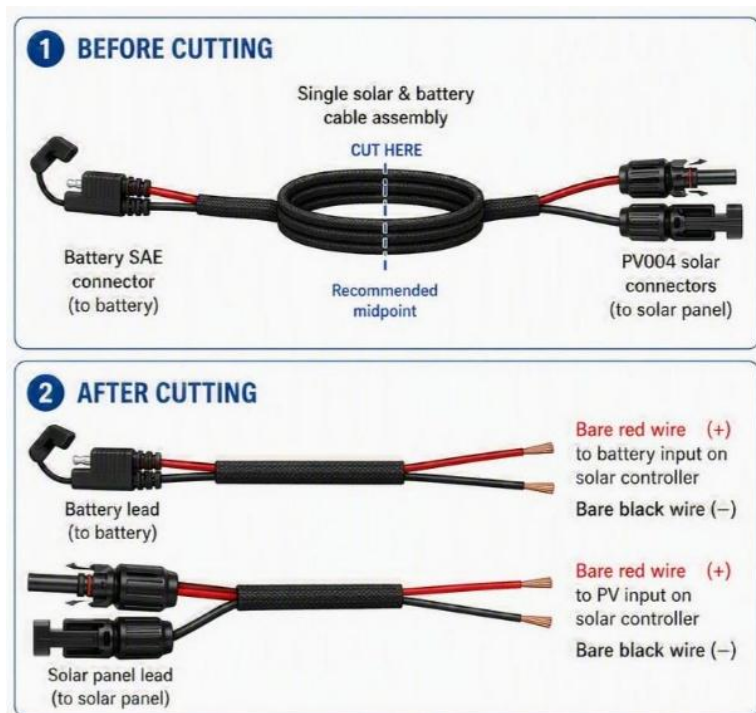
After cutting, the cable assembly will become two separate leads:

- **Solar panel lead:** one end has the PV004 solar panel connectors, and the other end has two bare wires.
- **Battery lead:** one end has the battery SAE connector, and the other end has two bare wires.

The bare wire ends from both leads will connect to the solar charge controller. Keep the red wire as positive and the black wire as negative when connecting to the controller.

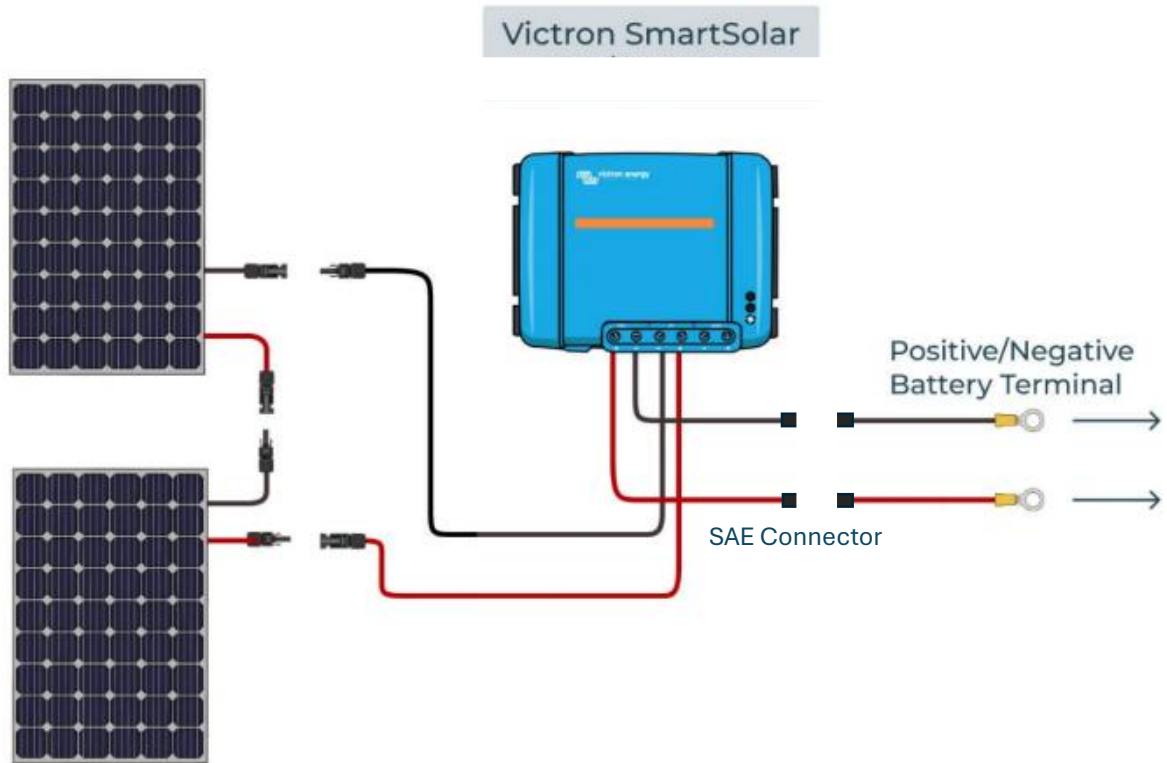
Installation Note:

Cut the cable only after confirming the approximate physical placement of the battery, solar panels, and main compute unit. Leave enough slack for routing, strain relief, and small position adjustments during final assembly.

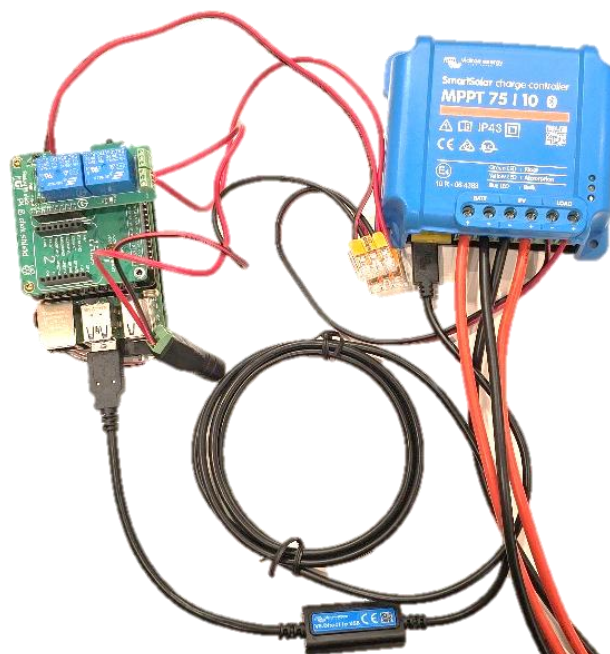


17. Install the cables onto the Victron SmartSolar as per the following diagram.
18. Install the battery terminal cables to SAE connector as per the following diagram.

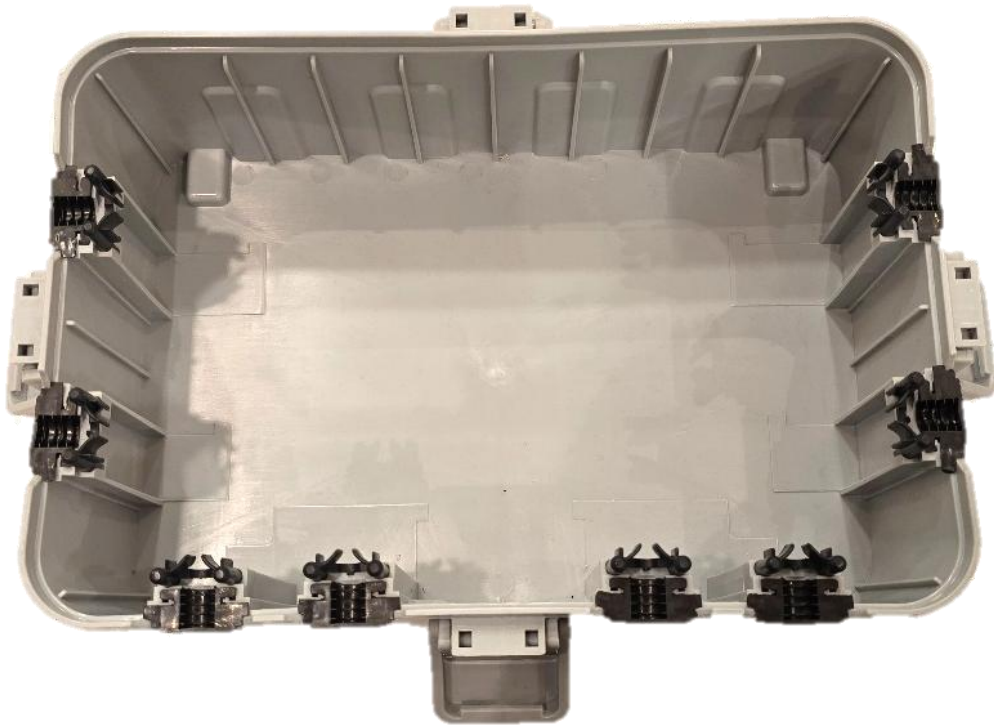
Important Note: The solar panels must be connected in **SERIES**. The connectors are polarized, meaning they can only go in one way. Closely follow the diagram below when making the connections. **DO NOT CONNECT THE CABLES THAT GO FROM THE SOLAR PANELS AND THE BATTERY TO THE CABLES WE HAVE INSTALLED INTO THE SOLAR CONTROLLER'S TERMINALS YET.**



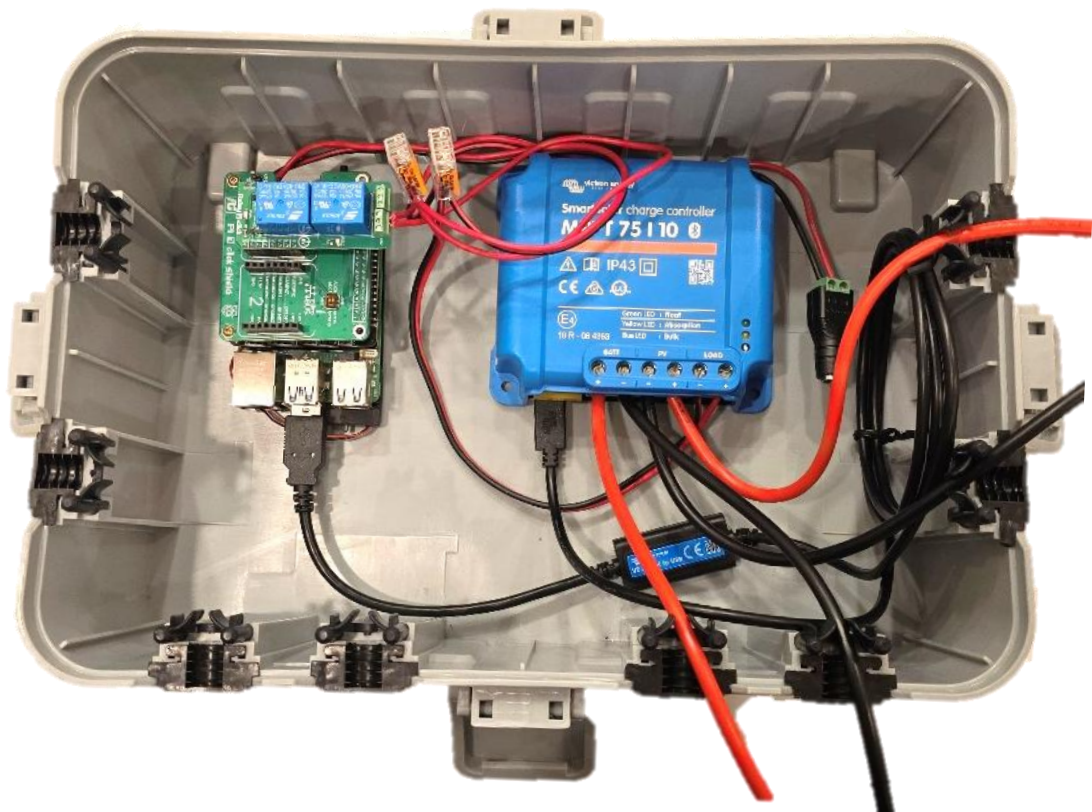
Install the **BLACK** for **Negative (-)** and **RED** for **Positive (+)** wires that go from the solar panels to the solar controllers PV positive and negative input respectively, following the previous diagram.



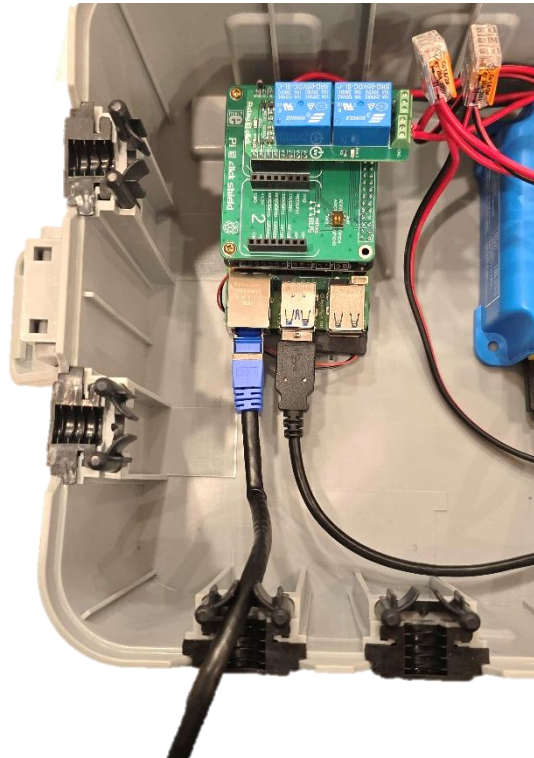
19. Prepare the weatherproof enclosure to start assembly.



20. Secure the Pi 5 assembly and the Solar controller using the 3M Dura Lock Velcro patches and position the components inside the case like pictured below, this will enable proper cable management.



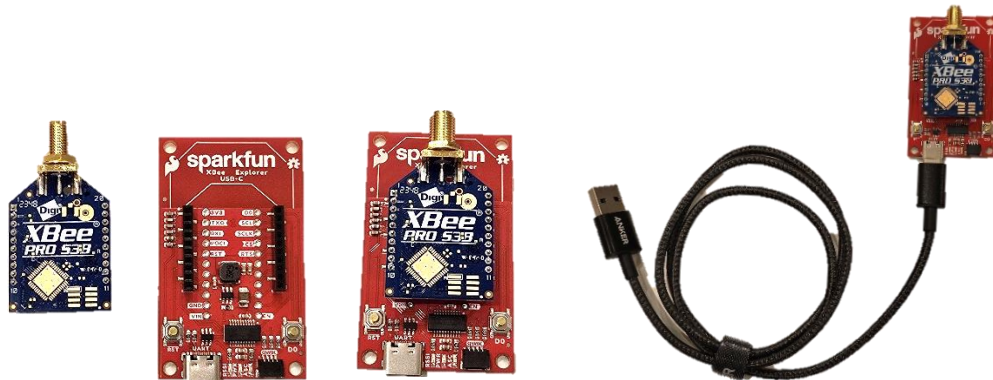
21. Connect the Starlink Ethernet cable, and route it through the rubber seal as shown below:



22. Connect the other end of the STARLINK ethernet cable to the weatherproof Starlink ethernet adaptor and assemble the weather-proof adaptor as described in the following diagram.



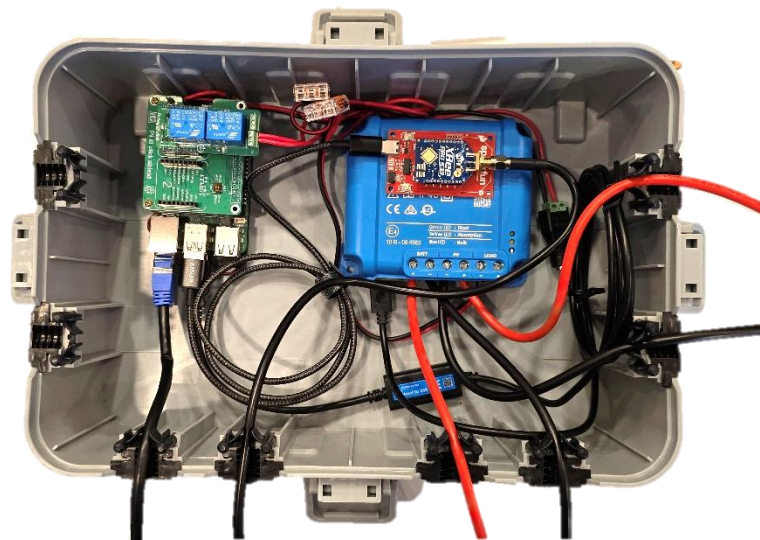
23. Connect the XBee board onto the USB-UART adapter board and to the USB-C cable as shown below, in this case we will use the USB-C version, as it allows you to freely position the board inside the enclosure.



24. Now connect the external XBee antenna using the RP-SMA antenna connector on the XBee chip:



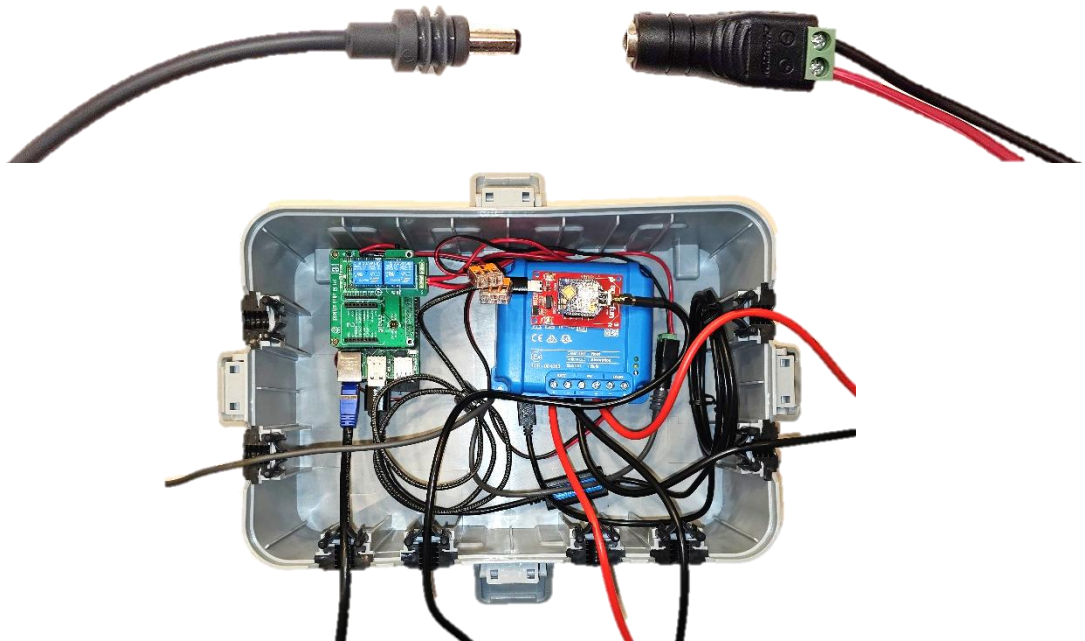
25. Connect the USB cable for the XBee boards into the Raspberry Pi 5 USB port and secure the XBee chip in place, you can use double sided tape or velcro, and attach it to the bottom of the case or like in this case to the solar controller:



26. Route the XBee antenna cable through one of the rubber seals to ensure a weatherproof seal once closed.

27. Finish off the build by installing the STARLINK DC Power Cable into the STARLINK DC female power plug, and route the cable out through one of the unused rubber seals on

the side to guarantee a weatherproof seal once closed



28. At this point if you are planning on running one of the optional AudioMoth devices for audio capture, you can connect the USB cable to the Raspberry Pi 5 and route the cable through the remaining rubber seal.
29. Connect the weather-proof ethernet connector to the STARLINK Mini ethernet port. Press it in and make sure the ethernet connector is correctly seated.
30. Connect the other end of the STARLINK POWER CABLE to the antenna. Press it in and make sure the terminal is fully seated. It is crucial for the power plug to be correctly seated inside the antennas power port to make sure it remains weather-proof.
31. **Double-check all previous connections** (AudioMoth, Xbee, case, etc.) are secure.
32. **Tighten all connections securely to avoid loose wiring.**
33. **Verify polarity before powering up**, incorrect wiring can damage the system.
34. **CONNECT THE CABLES THAT GO FROM THE SOLAR CONTROLLER TO THE SOLAR PANELS PV004 CONNECTORS AND FROM THE SOLAR CONTROLLER TO THE BATTERY SAE CONNECTOR.**
35. Once everything is connected correctly, the system should power on automatically.
36. Check the solar controller display or indicator lights to confirm power on status.
37. **CONGRATULATIONS! You have finished the main unit build assembly.**

7 Initial Raspberry Pi Configuration

After finishing the hardware build, perform the initial setup on your Raspberry Pi.

7.1 First Boot and Basic Setup

1. Connect Peripherals:

- **Keyboard and Mouse:** Connect via USB ports.
- **Monitor:** Connect using the HDMI port.
- **Network Connection:** Ensure your Raspberry Pi is connected to the internet via Ethernet or StarLink.

2. Power On:

- If not already powered on, connect the power supply to the Raspberry Pi.

7.2 Registering for a SPARROW Account and Obtaining an Access Key

1. Create an account on the SPARROW site:

Note: The steps in this Section 7.2 are for users who choose to sign-up for an account for access to and use of our SPARROW Studio dashboard which is governed by separate terms and conditions. Users choosing to host their own dashboard will need to register and obtain an access key from their own site or alter this requirement by modifying the client source code.

- Navigate to www.sparrowstudio.azure.com and click “Sign Up” or “Login”.
- To obtain an API click on an existing SPARROW project or create a new one (multiple SPARROW’s can be assigned to a project).

 + New Project

- On the edit project page select project type SPARROW:

Project type

Choose how images will enter this project.

Type

Sparrow — edge device (images/audio via API key)

Project type cannot be changed after creation.
Real-time projects use Inbox ingestion. Non-real-time projects use manual uploads. Sparrow projects use device API keys for edge uploads.

- Click generate device key. This will create the API Key for your new SPARROW device, this key is only shown once.

Pending onboarding keys

Generate a key here, then bring up a Sparrow unit with that key plus your account email (chalmerscar1@microsoft.com). The unit's first /v1/onboarding call consumes the key — at that point the device + its API key move into **Onboarded devices** above. The same key value serves both onboarding and uploads; there is no second secret.

 + Generate device key

No unconsumed onboarding keys.

- You will be prompted during the installation process of the Setup Script to enter the API Key you just generated and your email address.

7.3 Transferring and Running the SPARROW Setup Script

With your Raspberry Pi up and running, the next step is to set up the environment required for the SPARROW project. This involves transferring and executing the provided `sparrow_setup.sh` script.

1. Obtain the Setup Script:

- Ensure you have the `sparrow_setup.sh` script available on your host PC.
- This script is typically provided via the SPARROW project repository (e.g., GitHub).

To access the SPARROW project repository please visit: aka.ms/sparrow

2. Navigate to Script Location:

```
cd ~/
```

3. Change Permissions to Make Executable:

```
sudo chmod +x sparrow_setup.sh
```

4. Execute the Script with Root Privileges:

- It's crucial to run the setup script with `sudo` to grant the necessary permissions for installing software and modifying system configurations:

```
sudo ./sparrow_setup.sh
```

5. Follow Interactive Prompts:

- The script will guide you through the setup process.

8. SENSOR SETUP

8.1 Configuring ReoLink Wi-Fi Cameras

Once the docker container is running (7.3 section completed) you can connect and configure the ReoLink cameras to transmit images to the SPARROW unit.

1. Download the ReoLink app:

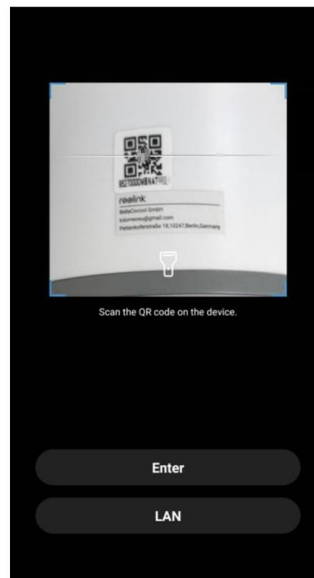
- download the ReoLink app on your mobile device from the Google Play Store or Apple App Store (no registration is required to use the ReoLink app):



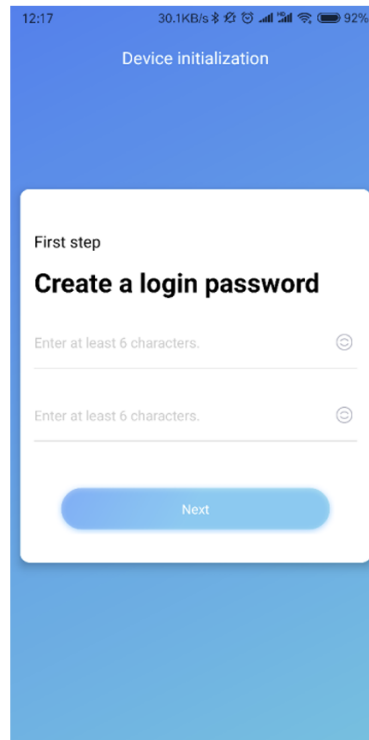
2. **⚠ IMPORTANT: Do NOT log in to any Reolink account in the app during setup.**
If you are already logged in, log out before continuing.
This step is critical—if you remain logged in, the cameras will not send alerts when there is no active internet connection.
3. **Connect your phone to the SPARROW Wi-Fi network:**
 - Open your Wi-Fi settings on your phone and search for the SPARROW SSID. Connect to the SSID and enter your password.
4. **Open the ReoLink app and click on the icon shown below to add a new camera:**



5. **Each camera comes with its own distinct QR code which can be found on the outside of the camera. Using the ReoLink app scan the QR code as shown below:**



6. **Follow the on-screen instructions to join the ReoLink camera to the SPARROW SSID.**
7. **Enter a local password:**



12:17 30.1KB/s 92%

Device initialization

First step

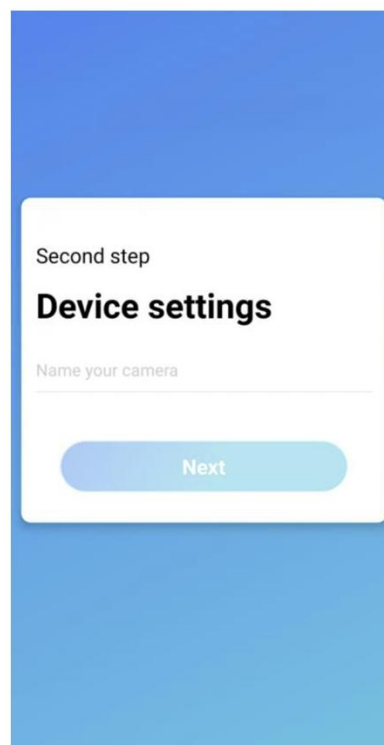
Create a login password

Enter at least 6 characters.

Enter at least 6 characters.

Next

8. Name the camera and click the button Next to enter the next page:



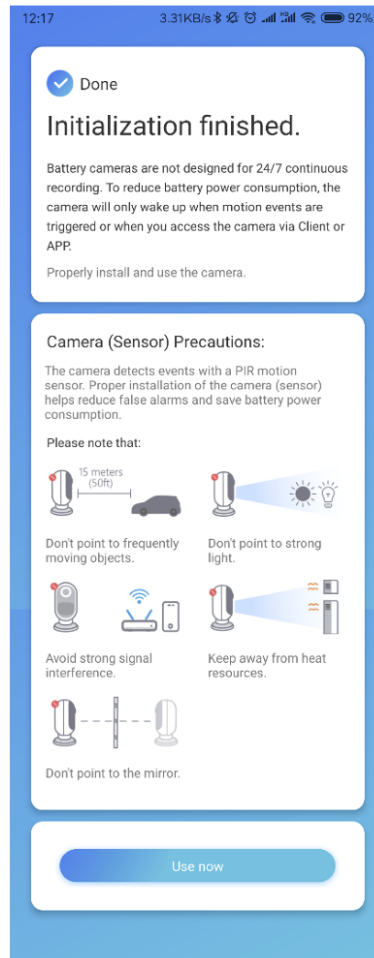
Second step

Device settings

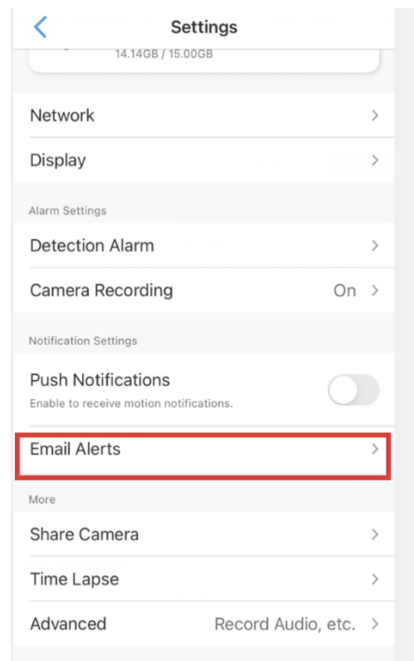
Name your camera

Next

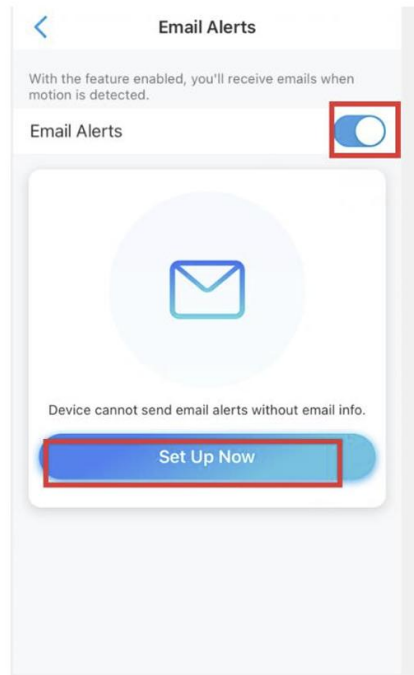
9. Read the tips and click Use now to finish the Initialisation:



10. On the camera settings page click on Email Alerts:



11. Enable Email Alerts then click Set up Now:



12. Fill in the required information and click save. You must use the following settings:

- Sender address: camera@localnetwork.com
- Password: (Check the sparrow setup script)
- SMTP Server 10.42.0.1
- Port: 25
- Turn off SSL/TLS
- Recipient Address 1: alerts@localnetwork.com (when using Android this setting is available after first setting it up with the Sender Address)
- Click Save
- Click Test and make sure you get the test success confirmation message

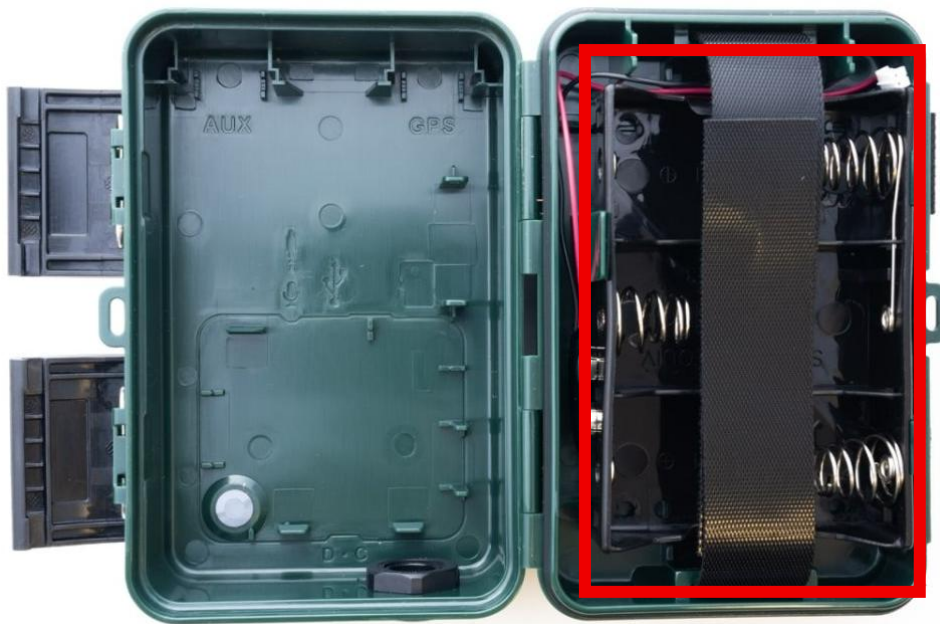
8.2 Peripheral Sensor Assembly (OPTIONAL)

AudioMoth DEV Microphone Setup

What You'll Need:

- AudioMoth DEV Board
- AudioMoth DEV Case
- Micro USB cable
- Cable gland (for the case)
- Precision knife (e.g., X-Acto knife)
- Internet access for firmware download

Step 1: Prepare the AudioMoth DEV Case.



1. **Open the case** and locate the battery cradle inside.
2. **Remove the battery cradle:**
 - a. Undo the straps holding it in place.
 - b. Lift the entire cradle out of the green box.
 - c. You should now have an empty case.

Step 2: Install the Cable Gland.



1. **Find the cable passthrough placeholder M16 cap on the case.**
2. **Remove the placeholder M16 cap.**
3. **Install a PG9 cable gland in its place.**
 - a. Make sure it is tightened securely to prevent movement.

Step 3: Prepare the Micro USB Cable.

1. **Check if the Micro USB connector fits through the cable gland.**
 - a. If it does not fit, proceed to the next step.
2. **Trim the Micro USB connector carefully:**
 - a. Use a precision knife to shave off some of the outer rubber casing.
 - b. Important: Do NOT cut into the internal data wires.
 - c. Take your time—small cuts are safer than deep cuts.

Step 4: Route the Cable.

1. **Insert the Micro USB cable through the cable gland** from the outside of the case.
2. **Pull enough cable inside** so it can comfortably reach the AudioMoth board.

Step 5: Connect and Position the AudioMoth.

1. **Plug the Micro USB connector into the AudioMoth.**
2. **Place the AudioMoth board inside the case** in its designated position.
3. **Ensure the cable is neatly routed** and not pinched by the case edges.

Step 6: Final Check.

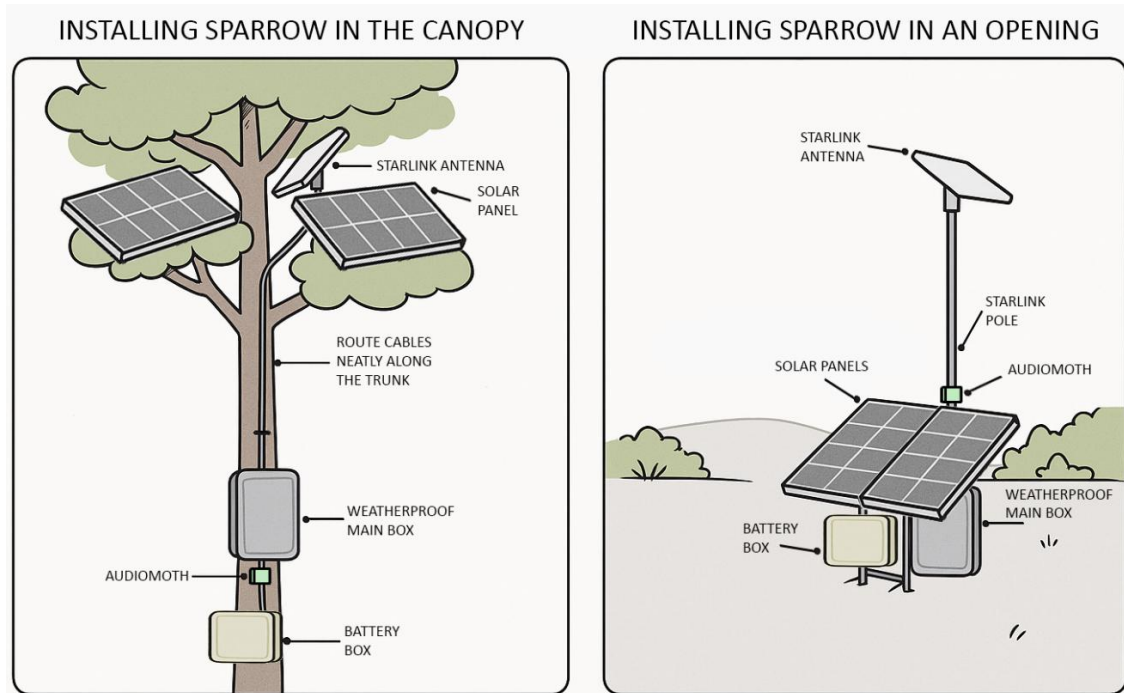
1. **Confirm the cable gland is tight.**
2. Make sure the AudioMoth is secure in its locking tabs, and the cable is connected properly.
3. Close the case.
4. AudioMoth is now ready for field deployment, use the provided Velcro straps or zip-ties to secure it in its final location.

9. Field Deployment Tips

Installing SPARROW in Tree Canopy

- When deploying SPARROW units in dense forest environments, mounting the system high in the tree canopy is critical for optimal performance.
- Position the solar panels and Starlink antenna at the highest possible point, ideally above or at the edge of the canopy, to minimize foliage interference. This ensures unobstructed line-of-sight for the antenna and maximum solar exposure throughout the day.
- Use secure climbing techniques or professional arborist support for safety and always verify that mounting hardware is rated for outdoor conditions and wind loads.
- Route all cables neatly along the trunk using UV-resistant cable ties and weatherproof junctions to prevent strain and moisture ingress.
- Finally, confirm that the installation is stable and that the solar panel angle is optimized for the local latitude to maximize energy capture.





Installing Sparrow in Treeline Openings (No Canopy Cover)

- For open-area deployments, Sparrow can be configured as a compact ground-based system.
- Use solar panel mounting brackets that elevate the panels off the ground and allow for adjustable tilt angles to optimize sunlight exposure year-round.
- Ensure the brackets are anchored securely to withstand wind and wildlife interference.
- The Starlink antenna should be mounted on a fixed pole support, positioned away from obstructions such as shrubs or terrain features to maintain a clear view of the sky.
- Keep all electronics housed in a weatherproof enclosure and elevate the enclosure slightly above ground level to avoid water pooling.
- For added reliability, consider using guy wires for pole stability and protective conduit for exposed cables.
- This setup is ideal for rapid deployment and easy maintenance in open landscapes.



Additional Suggestions

- **Weatherproofing:** Apply silicone seals or gaskets on all enclosure entry points to prevent moisture ingress.
- **Wildlife Protection:** Use mesh guards, zip-ties or double sheath wires to protect cables from rodents and birds.
- **Maintenance Access:** Leave enough slack in cables and plan for safe access to solar panels and antennas for cleaning or adjustments.
- **Power Monitoring:** If possible, integrate a remote monitoring solution to track solar performance and battery health.
- **AudioMoth Positioning:** The AudioMoth microphone can be secured under the main box to the tree trunk or the main cable bundle when installing SPARROW in the Canopy. For Opening installations, the AudioMoth recorder can be secured to the main solar panel mounting brackets.
- **Battery Case:** For deployments of SPARROW units in the wild, it is essential to house the battery system in a weatherproof container to extend battery life and overall lifespan. Using plastic totes or weatherproof bin containers provides effective protection against moisture and environmental exposure. To maintain the enclosure's integrity while routing cables, cable glands should be used to seal entry points and prevent water ingress. In cold environments, adding thermal insulation inside the container will help maintain optimal battery temperature and improve performance. This setup is especially beneficial in humid or flood-prone areas, where elevated placement and optional ventilation valves can further enhance durability.

10. SPARROW Unit Product Specification Sheet

Key Features & Capabilities

- **AI-powered edge computing** for wildlife and environmental monitoring
- **Solar-powered, autonomous operation** in remote locations
- **Multi-sensor integration:** acoustic, visual, environmental, and wireless sensors
- **Real-time data transmission** via low-Earth orbit satellites (Starlink), cellular (GSM, 3G, 4G, 5G)* or RF (XBee, LoRa)*
- **Robust field deployment options:** canopy or ground, weatherproof, wildlife-resistant

Component	Specification / Description	Details / Features
Main Compute Device	Raspberry Pi 5 w/8GB Ram	Edge AI processing, runs PyTorch-based wildlife models, interfaces with all sensors
Storage	M.2 NVMe SSD (PCIe Gen 3)	Stores OS and all logged data locally, 1TB or larger recommended
Power Supply	Solar 24V Power Matrix	Solar-powered, supports LiFePO4 or AGM batteries in 24V configuration
Solar Panel Array	2x 12V panels in series (24V array)	Powers system, maximizes solar exposure, overall dimensioning subject to use case
Battery Bank	LiFePO4 preferred, AGM compatible	24V 100Ah capacity recommended (a pair of 12V batteries connected in series can be used)
Solar Charge Controller	Victron SmartSolar 100/20 MPPT Controller	Manages charging, power profiles, and system voltage
Networking	Starlink Mini Satellite Internet	Location and power usage stats, advanced logging via Starlink app.
Wi-Fi Module	Onboard Wi-Fi card	For local wireless connectivity and remote monitoring
Sensor Expansion	Shuttle Click Expansion Board, BUS Board, Pi 3 Click Shield	mikroBUS standard sockets for stacking up to four click boards each, GPIO expansion hub
Environmental Sensors	Environment 3 Click Board (BME688)	Pressure: 300 to 110hPa Temperature: 0 to 65°C (±1°C) Humidity: 10-90%RH VOCs, VSCs, CO, H ₂ detection.
Real-Time Clock	Raspberry Pi 5 Internal RTC	Maintains accurate time during power interruptions
Wireless Connectivity	Digi XBee RF module	Low-power CE/RED RF module with superior performance and interference immunity from Digi International. The module can run either a proprietary DigiMesh or point-to-multipoint networking protocol. Operating between 863MHz and 870MHz (868MHz) or 902 MHz and 928MHz frequency range (900MHz) depending on location.
Relay Board	Relay 3 Click Board (SRD-5VDC-SL-C relays)	Dual relay, remote ON/OFF for Starlink, 28VDC/7A max load
Audio Sensor	AudioMoth DEV Board & Case (Optional)	USB Microphone, waterproof enclosure, firmware upgradable and remotely configurable.
Cameras	ReoLink Argus Eco Wi-Fi Camera	Temperature Range: -10 °C to 55 °C Weather Resistance: Rated IP65 Resolution: 1080p HD (1920 × 1080) Night Vision: Up to 10 meters (33 feet) Wi-Fi: IEEE 802.11 b/g/n Battery: 5200mAh rechargeable battery Solar: Compatible with Reolink Solar Panel Configurable via app, QR code setup, email alerts, SMTP integration.

*Support for these communication systems pending development.

11. Weatherproof Enclosure Build Instructions for Tough Conditions (OPTIONAL)

1. Install all components to the removable grid backplate and secure it to the inside of the electrical junction box using the provided screws. Carefully flex the black plastic backplate to force the backplate past the protruding vents inside the electrical junction box.
2. Once the backplate is secure in place, using a conical stepper bit, drill on the right side of the bottom of the case, two 25mm (0.98 inches) holes.



38. Prepare the battery and solar cable leads

Before installation, decide how much cable length is needed between the main compute unit, the solar panels, and the battery. The cable assembly can be cut to match the installation layout.

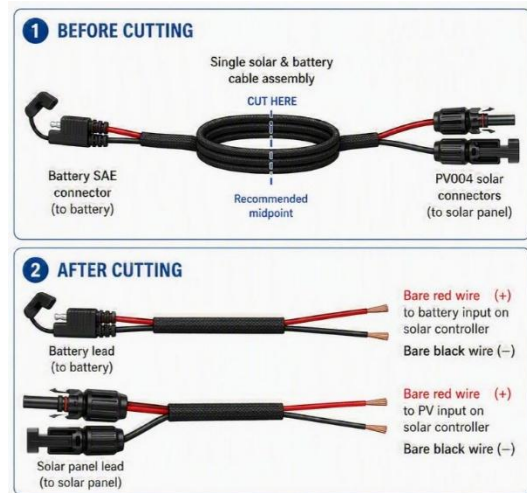


We recommend cutting the cable assembly in half. This gives enough length on both sides to position the solar panels and battery with some flexibility during installation.

After cutting, the cable assembly will become two separate leads:

- **Solar panel lead:** one end has the PV004 solar panel connectors, and the other end has two bare wires.
- **Battery lead:** one end has the battery SAE connector, and the other end has two bare wires.

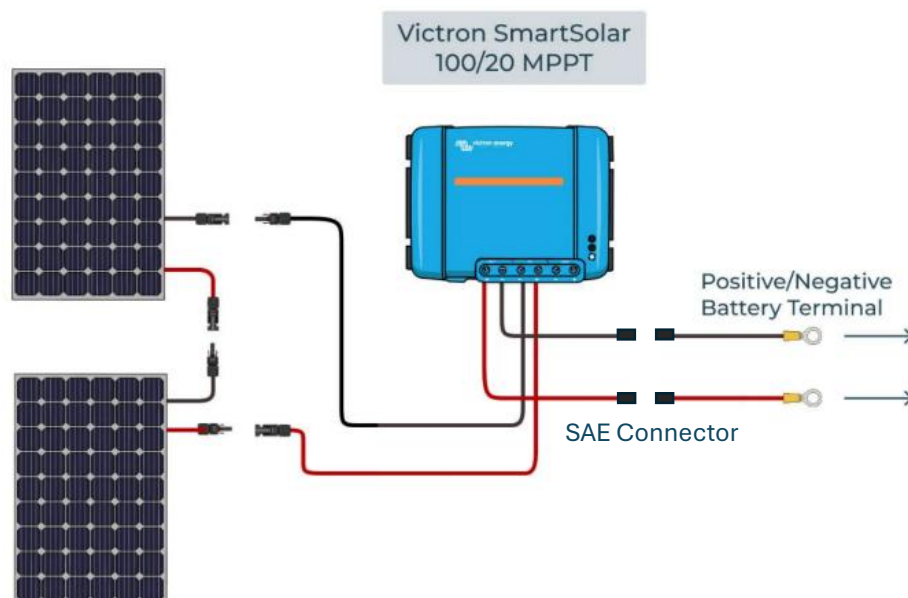
The bare wire ends from both leads will connect to the solar charge controller. Keep the red wire as positive and the black wire as negative when connecting to the controller.



Installation Note:

Cut the cable only after confirming the approximate physical placement of the battery, solar panels, and main compute unit. Leave enough slack for routing, strain relief, and small position adjustments during final assembly.

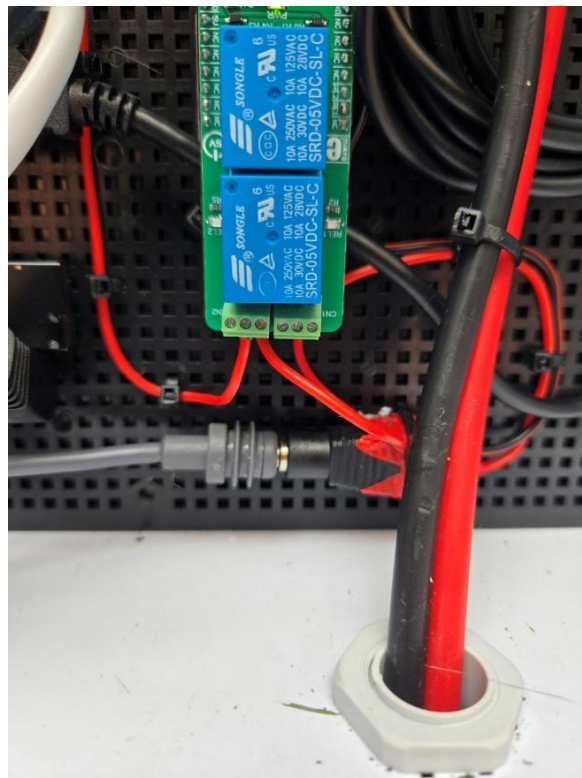
39. Install the cables onto the Victron Solar Controller as per the following diagram.
40. Install the battery terminal cables to SAE connector as per the following diagram.
Important Note: The solar panels must be connected in **SERIES**. The connectors are polarized, meaning they can only go in one way. Closely follow the diagram below when making the connections. **DO NOT CONNECT THE CABLES THAT GO FROM THE SOLAR PANELS AND THE BATTERY TO THE CABLES WE HAVE INSTALLED INTO THE SOLAR CONTROLLER'S TERMINALS YET.**



3. Install the **BLACK** for Negative (-) and **RED** for Positive (+) wires that go from the solar panels to the solar controllers PV positive and negative input respectively, following the previous diagram.

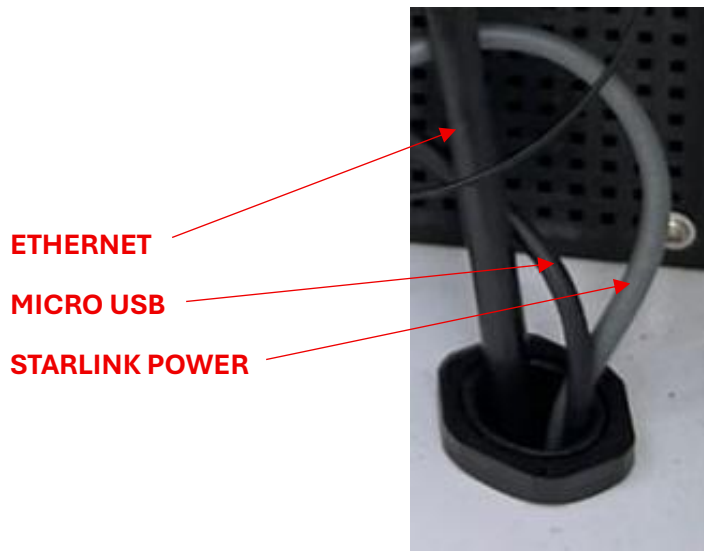


*Please note that the **Orange and Blue** wires shown in this image are the cables that go to the battery, these wires will be **BLACK** for Negative (-) and **RED** for Positive (+) respectively when using the battery cable listed and recommended in the BOM.



4. Use the **SECOND RIGHTMOST** 25mm (0.98 inches) hole and install the other PG19 gland.

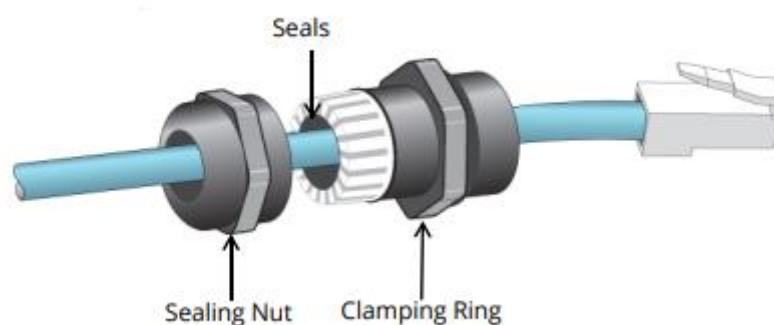
5. Route through the cable gland, the STARLINK Ethernet Cable, STARLINK POWER cable and the Micro USB cable for the AudioMoth as shown in the image below.



To achieve this:

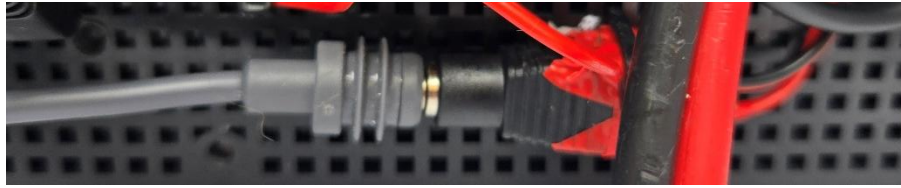
- a. First, route the STARLINK POWER cable, through the Sealing Nut of the cable gland, and then insert it through the Clamping Ring and into the Case.
- b. Second, route the micro-USB cable for the AudioMoth, this cable must be inserted and routed from the inside of the case. Route the micro-USB connector outward, through the cable gland and Sealing Nut. The USB type A connector will remain inside the case.
- c. Finally insert the ethernet cable following the diagram below. **If you are using an ethernet cable with a protective sheath over the connector, trimming or removal of the protective sheath might be necessary to fit through the gland.**

The following diagram shows how the cables should be routed through the cable gland.



6. Connect the AudioMoth USB cable and the STARLINK Ethernet cable to their respective USB (you can use any port) and Ethernet port on the Raspberry Pi 5.
7. **IMPORTANT NOTE:** To ensure proper weather-tightness of all cable entries, verify that **cable glands are fully seated, correctly assembled, and securely tightened.** Improper installation may lead to water ingress and damage to internal components.

8. Connect the STARLINK POWER cable to its corresponding female connector.



9. **When using the external 10dB long range optional fiberglass XBee antenna:** With the stepper bit and drill, make one 16mm (0.6 inches) on the left bottom side of the case.

Install the smaller PG9 cable glands on the pair of new 16mm (0.6 inches) holes on the box and use them to route the external antenna cables into the electrical Junction box.

RP-SMA

MALE

FEMALE



When using the 6dB common plastic Wi-Fi antenna: continue by drilling 2 small holes on the side of the case to install the FEMALE RP-SMA connector by securing it in place with the provided lock nut as shown below:



10. Connect the external antennas or antenna cables to the Wi-Fi RP-SMA connectors.
11. Connect the other end of the STARLINK ethernet cable to the weatherproof Starlink ethernet adaptor and assemble the weather-proof adaptor as described in the following diagram.



12. Connect the weather-proof ethernet connector to the STARLINK ethernet port. Press it in and make sure the ethernet connector is correctly seated.
13. Connect the other end of the STARLINK POWER CABLE to the antenna. Press it in and make sure the terminal is fully seated. It is crucial for the power plug to be correctly seated inside the antennas power port to make sure it remains weather-proof.
14. **Double-check all previous connections** (AudioMoth, Raspberry Pi 5, case, etc.) are secure.
15. **Tighten all connections securely to avoid loose wiring.**
16. **Verify polarity before powering up**, incorrect wiring can damage the system.
17. **CONNECT THE CABLES THAT GO FROM THE SOLAR CONTROLLER TO THE SOLAR PANELS PV004 CONNECTORS AND FROM THE SOLAR CONTROLLER TO THE BATTERY SAE CONNECTOR.**
18. Once everything is connected correctly, the system should power on automatically.
19. Check the solar controller display or indicator lights to confirm power on status.
20. **CONGRATULATIONS! You have finished the main unit build assembly.**